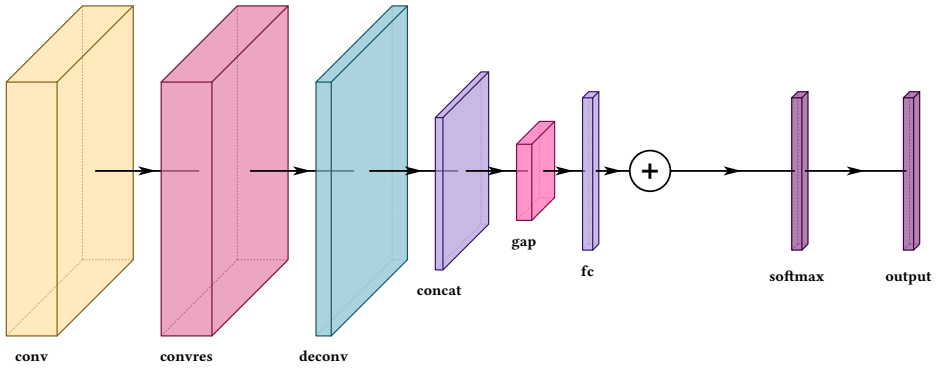
VINNT

Vision Illustration of Neural Networks in Typst



Version 0.1.0 · J.C. Vaught

Every figure in this manual is compiled from the code printed beside it.

CHAPTER 0

Contents

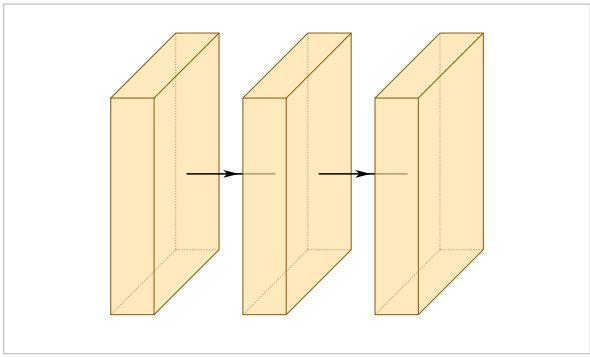
CHAPTER 1

What this is

VINNT draws layered neural network architectures as isometric block diagrams, the kind that open a paper's method section. It is a Typst package built on CeTZ.

You describe the network as an array of layers. The package works out the geometry, the spacing, the arrows between blocks, the routing of any skip connection you add, and the legend.

```
#draw-network((
  conv(),
  conv(),
  conv(),
))
```



Three convolutions, no options at all. Sizes, spacing and arrows are defaults.

1.1 How this manual is organised

Every option is shown on its own, usually across several figures, because one picture of an option tells you it exists and three tell you what it does. Where an option has a failure mode, the failure is shown next to the fix. Each chapter ends with a composite figure built from what the chapter covered, and chapter 15 is nothing but composites.

Complete published architectures — AlexNet, VGG, ResNet, U-Net, FCN-8, YOLO26-n and six RGB-IR fusion variants — are not here. They are whole readable files in `examples/` in the repository, and they are better read as files than quoted as fragments.

1.2 Installing

```
#import "@preview/vinnt:0.1.0": draw-network, conv, pool, input
```

Every example in this manual assumes an import like that one. Import `draw-network` plus whichever constructors you use, or take everything with `: *`, which is what the examples here do.

> VINNT is not yet published on Typst Universe. Until it is, point the import at a local copy: `#import "path/to/src/lib.typ": *`, which is what every example in this manual does.

1.3 The shape of a figure

```
#draw-network(
  layers,
  connections: (),
  groups: (),
  ...
)
```

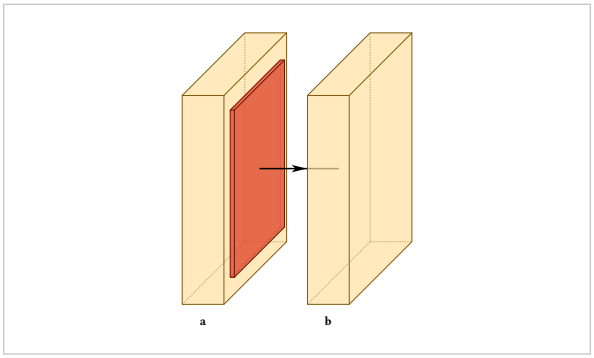
Everything in this manual is either an option on a layer, an option on a connection or a group, or one of those figure-wide settings.

CHAPTER 2

Writing a layer

A layer is a constructor call. Every type has one, named after it, and its arguments are the options:

```
#draw-network((
  conv(label: "a"),
  pool(),
  conv(label: "b"),
))
```



A constructor's signature is exactly the set of options that type accepts, so an editor can list them while you type, and Typst rejects a misspelled argument by name before the package ever sees it. A plain dictionary in the layer list is an error rather than a second way to write the same thing, so every figure reads the same. When options arrive as a dictionary anyway — shared settings, an imported dump — spread it into the constructor: `conv(..opts)`.

2.1 Options that do not exist are errors

An option a layer type does not read is rejected, not ignored:

```
error: unknown layer option "hieght" on layer 3 (type "conv").
Did you mean "height"? Options accepted here: bandfill, channels,
connection-label, depth, fill, height, image, label, ...
```

This matters more than it sounds. A key that is quietly ignored produces a figure that is merely wrong, and a wrong figure reads as the package being broken rather than as the typo it is. The check covers layer options, connection options, group options, unknown layer types, a layer with no `type`, and a connection or group naming a layer that has no such `name`.

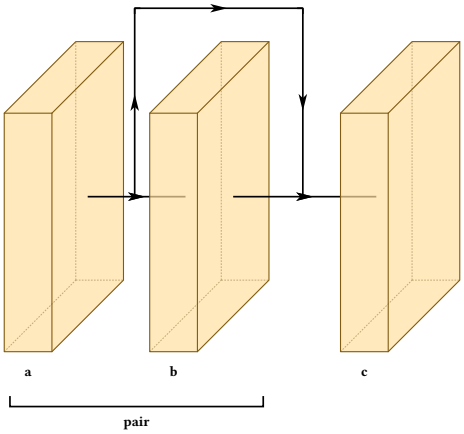
> The suggestion searches the options that type actually accepts, so `widths` on a `pool` is caught as readily as `hieght` on a `conv`. The first is spelled correctly and still means nothing there.

2.2 Naming a layer

`name` default `none`

An identifier for this layer. Never printed. Connections and groups find a layer by it, and nothing else does, so a layer with no name cannot be referred to.

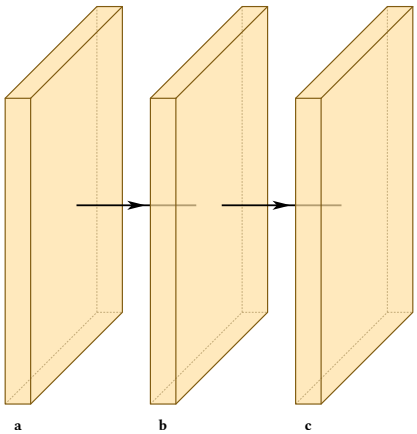
```
#draw-network(
  (
    conv(name: "a", label: "a"),
    conv(name: "b", label: "b"),
    conv(name: "c", label: "c"),
  ),
  connections: (connection(from: "a", to: "c"),),
  groups: (group(from: "a", to: "b", label: "pair"),),
)
```



2.3 The list is just an array

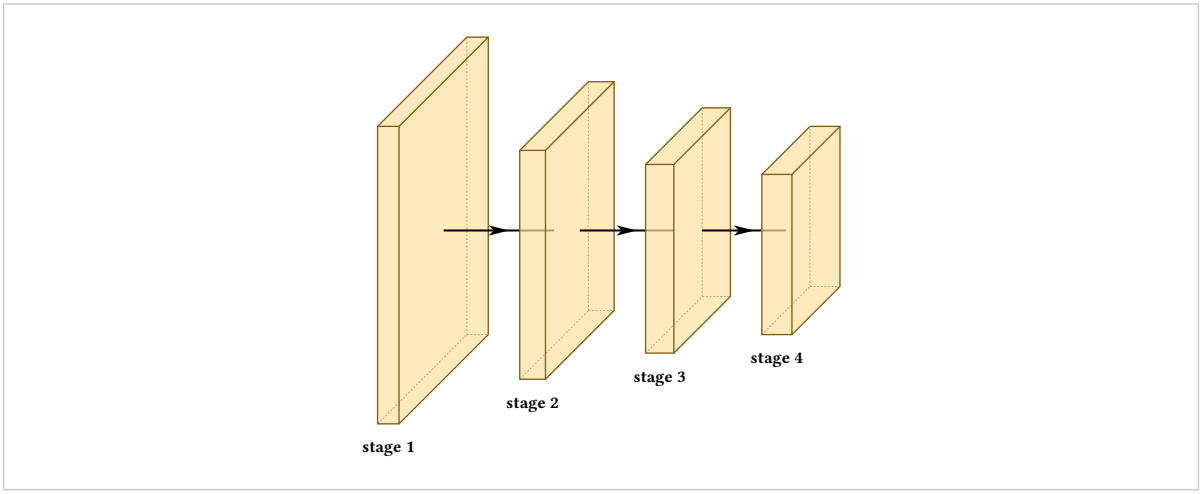
Nothing requires you to write layers out one by one. The list is an array and a constructor is a function, so ordinary Typst builds both. Sharing options across several layers is a spread:

```
#let big = (height: 6, depth: 6, widths: (0.5,))

#draw-network((
  conv(label: "a", ..big),
  conv(label: "b", ..big),
  conv(label: "c", ..big),
))
```



And generating layers is a `map`:

```
#let stage(n) = conv(
  label: "stage " + str(n),
  shape: (32 * n, 128 / n, 128 / n),
)

#draw-network(range(1, 5).map(stage))
```



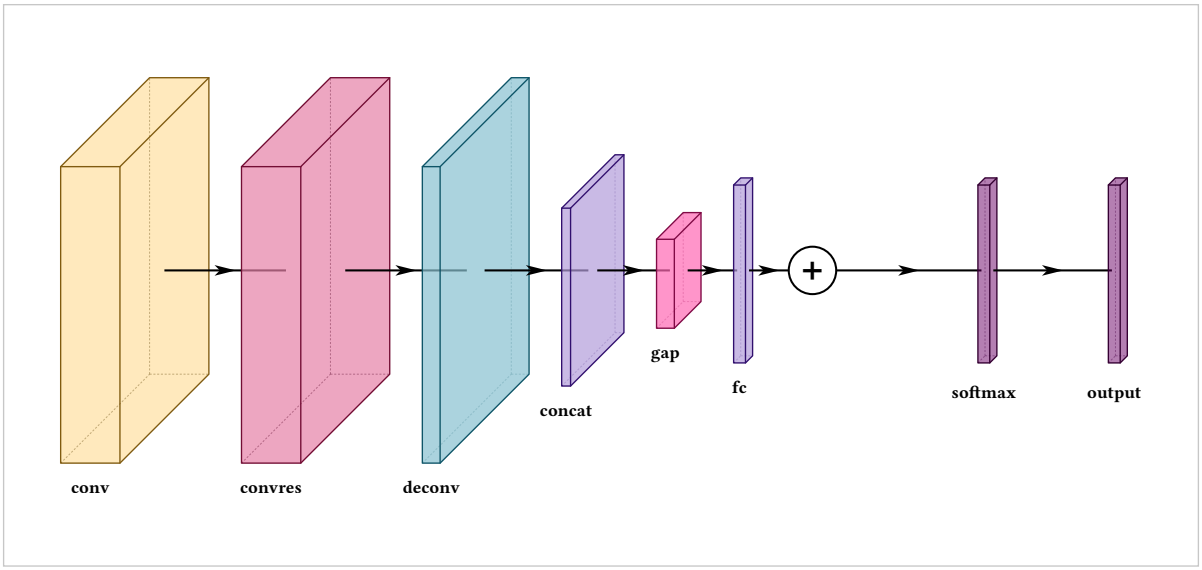
This is the difference between a figure you maintain and a figure you rewrite. Section 15.5 takes it further.

CHAPTER 3

Layer types

Fourteen types are predefined. Each carries a colour, a default size, a default legend entry, and in three cases a behaviour.

```
#draw-network((
  conv(label: "conv"),
  convres(label: "convres"),
  deconv(label: "deconv"),
  concat(label: "concat"),
  gap(label: "gap"),
  fc(label: "fc"),
  sum(),
  softmax(label: "softmax"),
  output(label: "output", offset: 2),
))
```



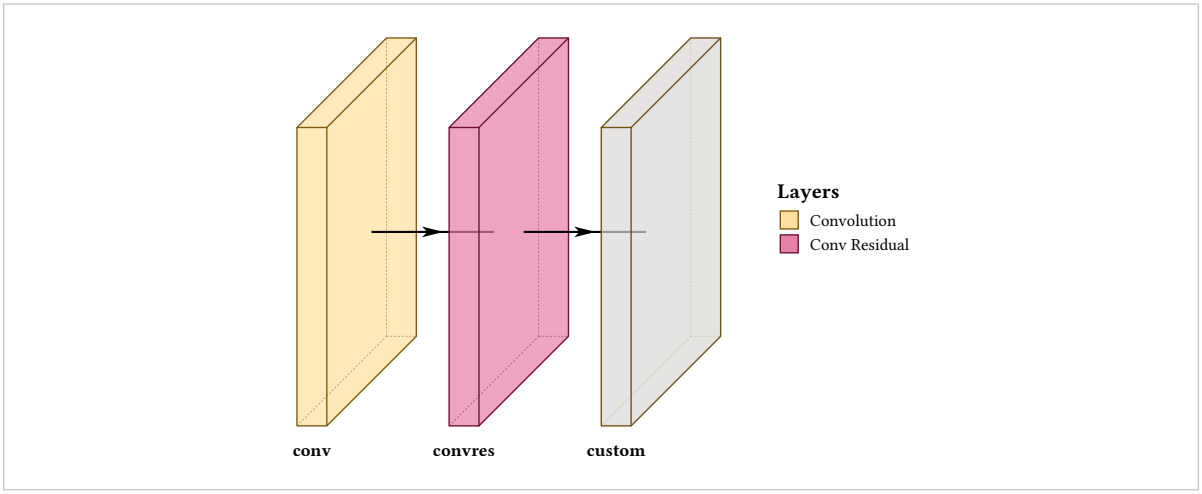
Also predefined: `input`, `pool`, `unpool`, `convsoftmax`, `custom`.

Type governs colour and defaults, not capability. Any type can be resized, recoloured, relabelled and given an image. Choose by what the block **means**: the type decides what the legend calls it and what colour a reader will come to associate with it across your figures.

3.1 The convolution family

`conv` and `convres` differ only in colour and legend text. `custom` is the same block with no meaning attached, and is the one you extend (chapter 9).

```
#draw-network(
  (
    conv(label: "conv", widths: (0.5,)),
    convres(label: "convres", widths: (0.5,)),
    custom(label: "custom", width: 0.5),
  ),
  show-legend: true,
)
```



3.2 Pooling attaches to the block in front

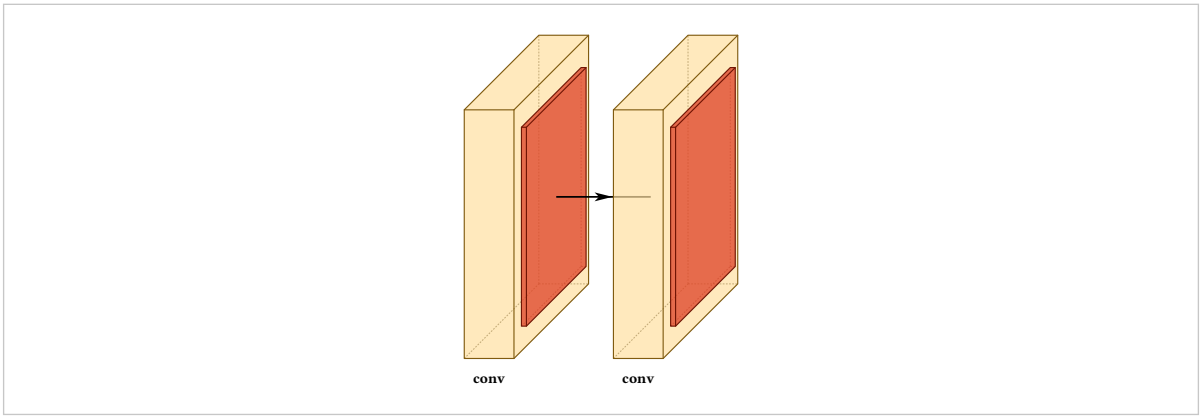
`offset`

default `auto`

read specially by `pool` and `unpool`

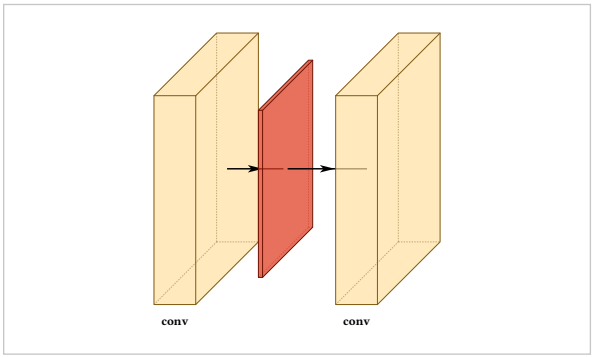
On `pool` and `unpool`, leaving `offset` unset attaches the layer to the block before it, with no arrow between them. That is what makes a convolution-plus-pool read as one stage rather than two.

```
#draw-network((
  conv(label: "conv"),
  pool(),
  conv(label: "conv"),
  pool(),
))
```



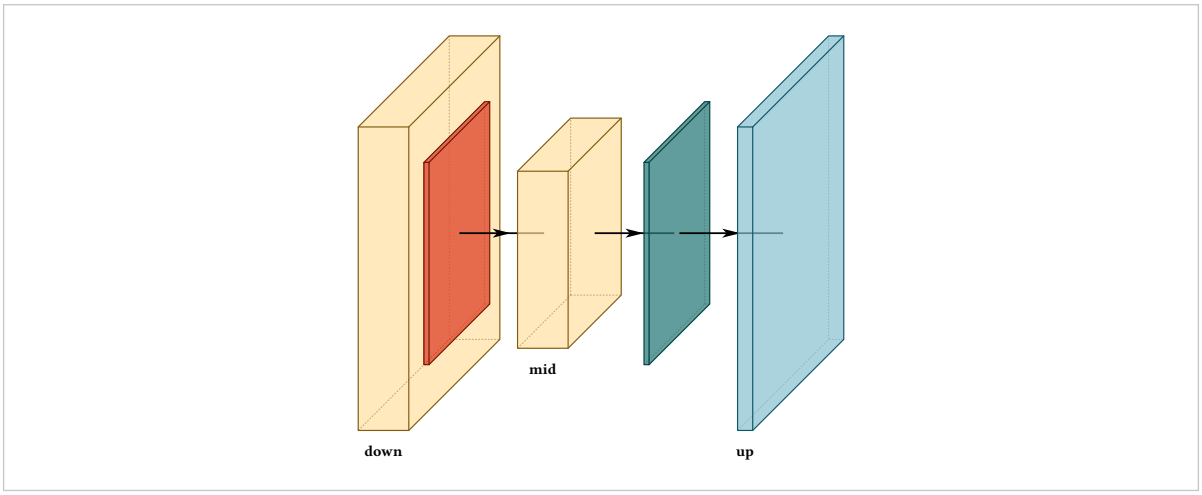
Give one an `offset` and it detaches, and the axis arrow appears:

```
#draw-network((
  conv(label: "conv"),
  pool(offset: 1.5),
  conv(label: "conv"),
))
```



An encoder-decoder uses both, `pool` on the way down and `unpool` on the way up:

```
#draw-network((
  conv(label: "down", height: 6, depth: 6),
  pool(height: 4, depth: 4),
  conv(label: "mid", height: 3.5, depth: 3.5),
  unpool(height: 4, depth: 4, offset: 1.5),
  deconv(label: "up", height: 6, depth: 6),
))
```



3.3 The sum node

`sum` is a node, not a block. It has a `radius` where other types have a width, and a `symbol` you can replace.

`radius` default `0.4`

`sum`

Size of the disc.

`symbol` default `+`

`sum`

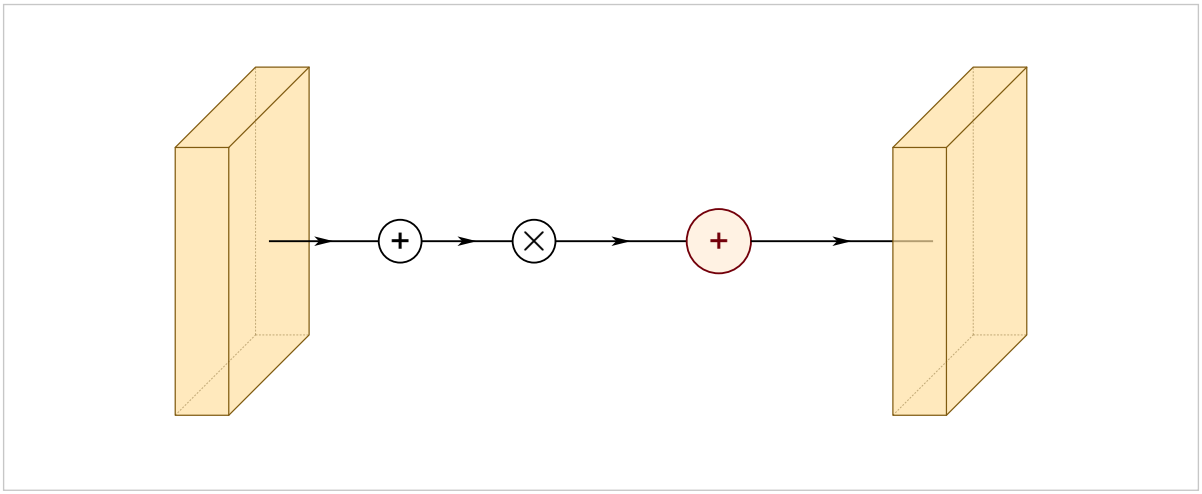
Any content, centred in the disc.

`stroke` default `black`

`sum`

Outline colour.

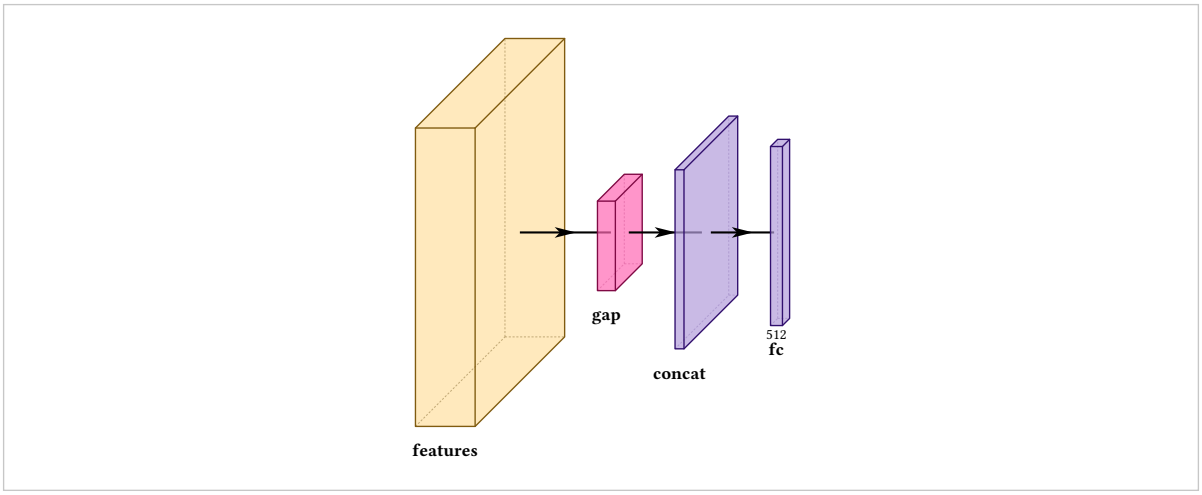
```
#draw-network((
  conv(),
  sum(),
  sum(symbol: [$times$]),
  sum(radius: 0.6, fill: rgb("#FFF2E3"), stroke: rgb("#73000A")),
  conv(),
))
```



3.4 Pooling to a vector, and joining

`gap` collapses a feature map to a vector and is small by default. `concat` joins two paths, and is the usual target for a fan of arriving routes (section 10.6).

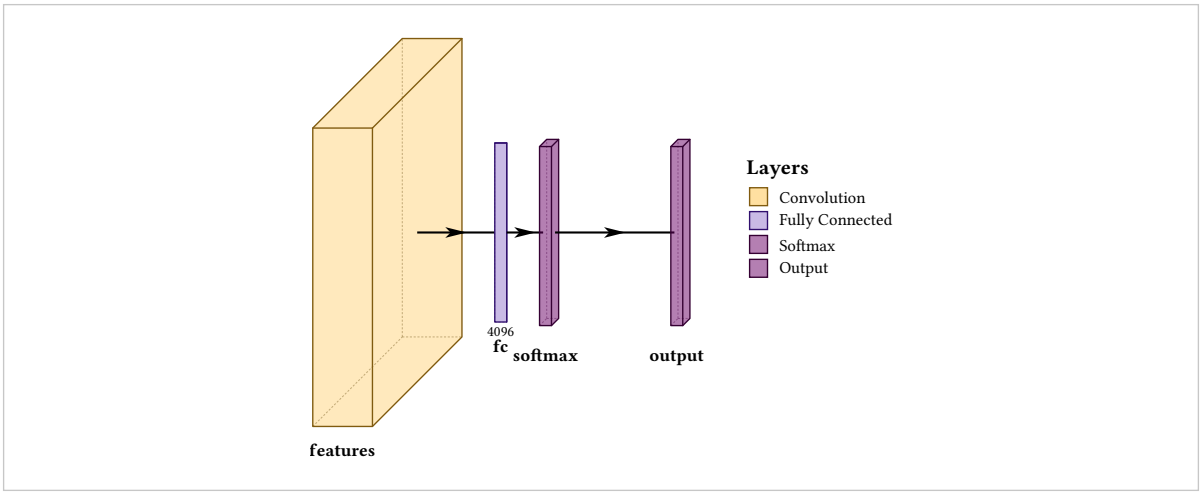
```
#draw-network((
  conv(label: "features", height: 5, depth: 5),
  gap(label: "gap"),
  concat(label: "concat"),
  fc(label: "fc", channels: (512,)),
))
```



3.5 Classifier heads

`fc`, `softmax`, `convsoftmax` and `output` are the tail of a classifier. `fc` with `depth: 0` draws as a flat rectangle, which is what a vector looks like.

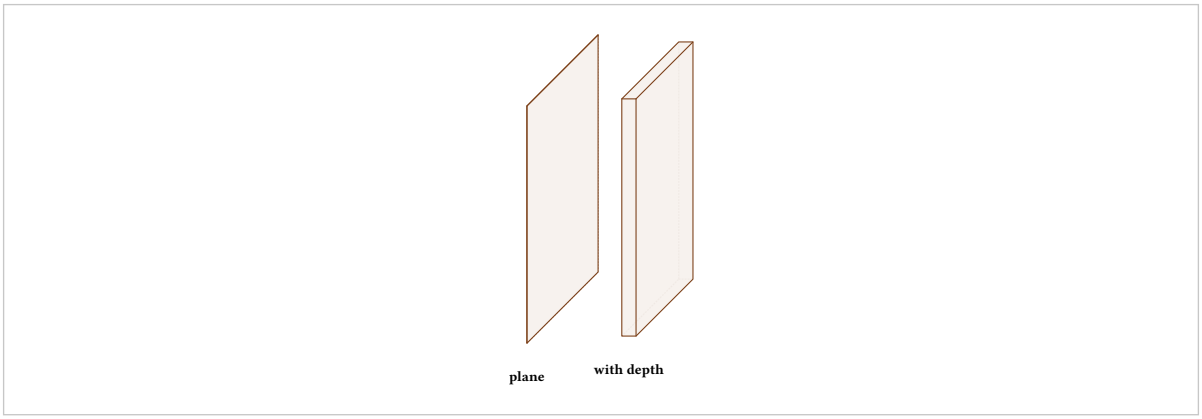
```
#draw-network(
  (
    conv(label: "features"),
    fc(label: "fc", channels: (4096,), depth: 0),
    softmax(label: "softmax"),
    output(label: "output", offset: 2),
  ),
  show-legend: true,
)
```



3.6 The input layer

`input` defaults to a flat plane with no thickness, which is what an image is. Give it a `width` and a `depth` to make it a block instead. It is also the one type with no outgoing arrow by default (section 10.9).

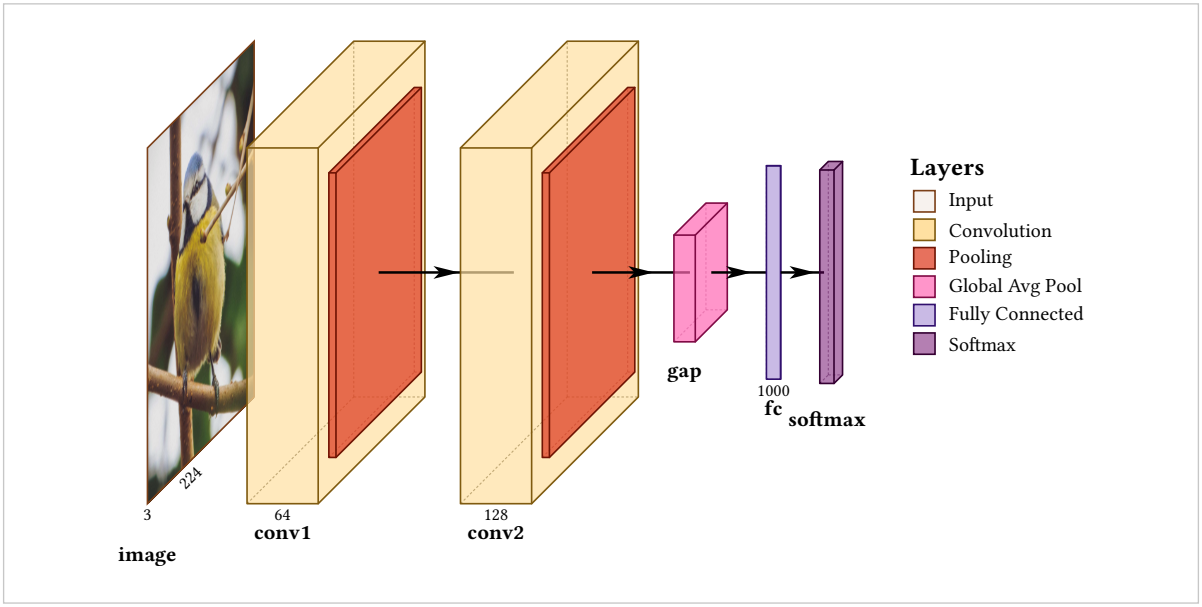
```
#draw-network((
  input(label: "plane"),
  input(label: "with depth", depth: 4, width: 0.3, offset: 2),
))
```



3.7 Putting the types together

Nothing below sets a size, a colour or a gap. It is types, labels and channel counts only, and it is already a readable figure.

```
#draw-network(
  (
    input(image: "default", label: "image", channels: (3, 224)),
    conv(label: "conv1", channels: (64,), offset: 1.4),
    pool(),
    conv(label: "conv2", channels: (128,)),
    pool(),
    gap(label: "gap"),
    fc(label: "fc", channels: (1000,), depth: 0),
    softmax(label: "softmax"),
  ),
  show-legend: true,
)
```



CHAPTER 4

Sizing a block

A block has three dimensions. `height` and `depth` are the spatial extent of the tensor; thickness is `width` or `widths`, and conventionally means channel count.

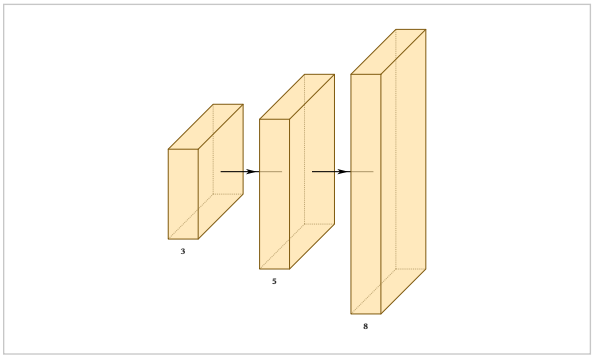
`height` default `5, varies by type`

Vertical extent, in canvas units.

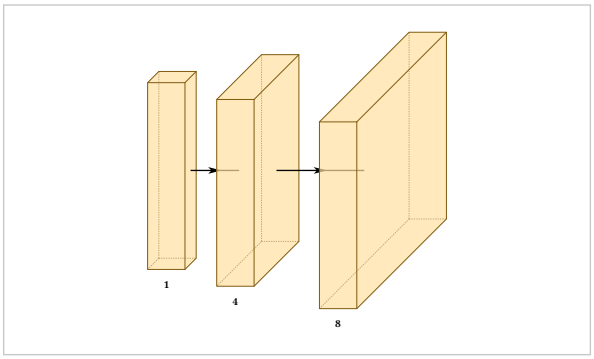
`depth` default `5, varies by type`

Extent along the projection. `0` draws a flat rectangle.

```
#draw-network((
  conv(label: "3", height: 3),
  conv(label: "5", height: 5),
  conv(label: "8", height: 8),
))
```




```
#draw-network((
  conv(label: "1", depth: 1),
  conv(label: "4", depth: 4),
  conv(label: "8", depth: 8),
))
```



4.1 Thickness

`width` default `varies`

`input`, `deconv`, `concat`, `convsoftmax`, `softmax`, `output`, `custom`

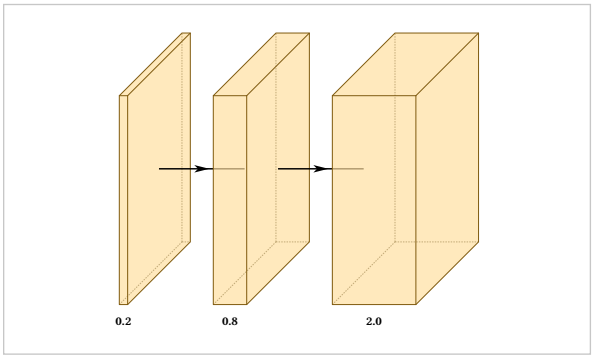
A single thickness.

`widths` default `(1,)`

`conv`, `convres`, `custom`

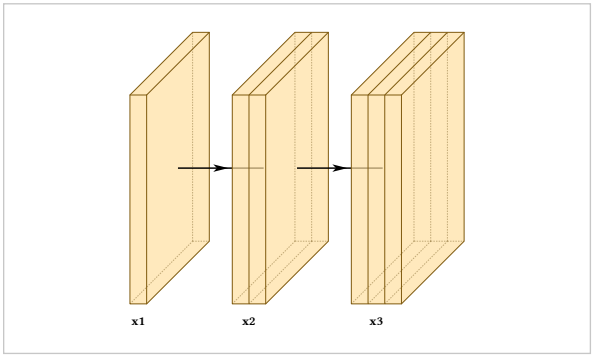
An array of thicknesses, drawing one band per entry inside a single block.

```
#draw-network((
  conv(label: "0.2", widths: (0.2,)),
  conv(label: "0.8", widths: (0.8,)),
  conv(label: "2.0", widths: (2.0,)),
))
```



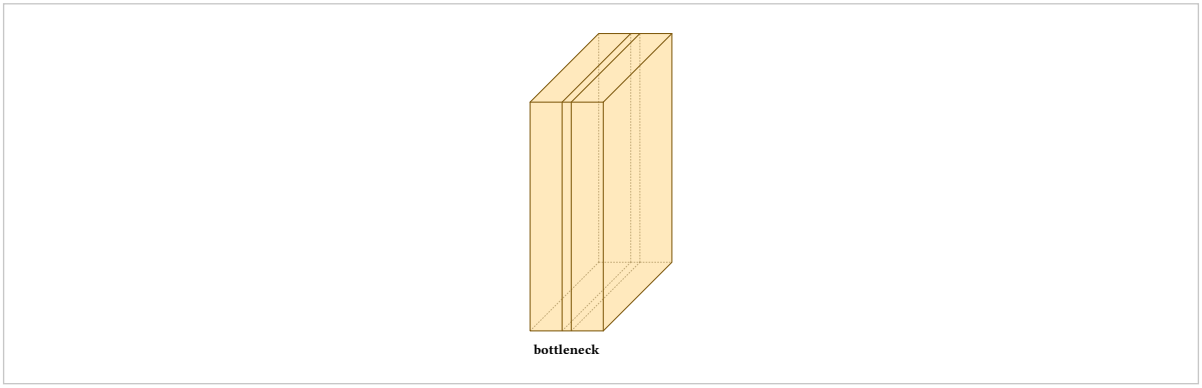
`widths` is how a stage of two or three convolutions at one resolution is normally drawn: one block, banded, rather than three blocks in a row.

```
#draw-network((
  conv(label: "x1", widths: (0.4,)),
  conv(label: "x2", widths: (0.4, 0.4)),
  conv(label: "x3", widths: (0.4, 0.4, 0.4)),
))
```



The bands need not be equal, which is how you draw a bottleneck:

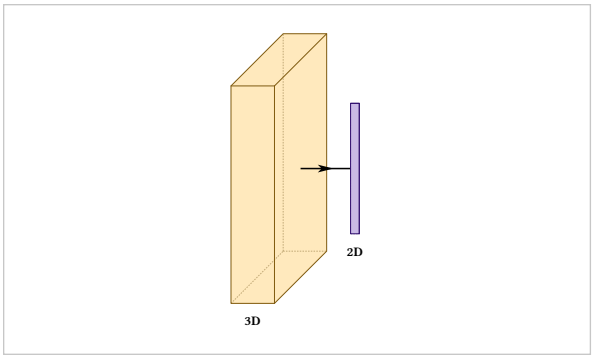
```
#draw-network((
  conv(label: "bottleneck", widths: (0.7, 0.2, 0.7)),
))
```



`width` and `widths` are different options and not interchangeable. No type accepts both except `custom`, and using the wrong one is an error naming the other.

4.2 Flat layers

```
#draw-network((
  conv(label: "3D", depth: 4),
  fc(label: "2D", depth: 0),
))
```



4.3 Sizing from the tensor shape

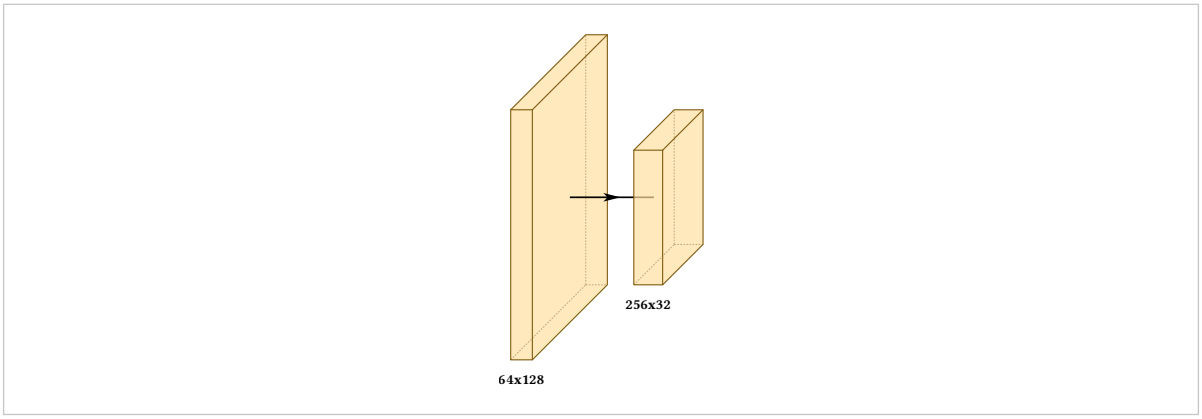
Stating three numbers per block and keeping them consistent across a pyramid is the tedious part of drawing one of these, and the part that rots when the model changes.

shape

default none

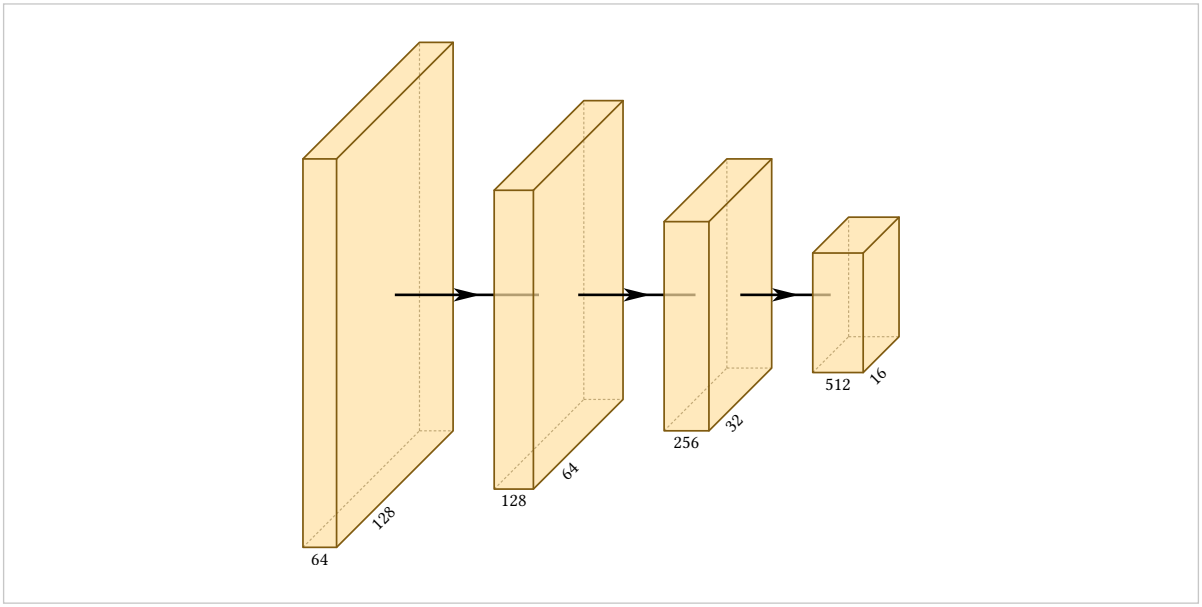
The tensor this layer produces, as `(channels, height, width)`. Height, depth and width are derived from it.

```
#draw-network((
  conv(shape: (64, 128, 128), label: "64x128"),
  conv(shape: (256, 32, 32), label: "256x32"),
))
```



Across a pyramid it is the whole geometry:

```
#draw-network((
  conv(shape: (64, 128, 128), channels: (64, 128)),
  conv(shape: (128, 64, 64), channels: (128, 64)),
  conv(shape: (256, 32, 32), channels: (256, 32)),
  conv(shape: (512, 16, 16), channels: (512, 16)),
))
```



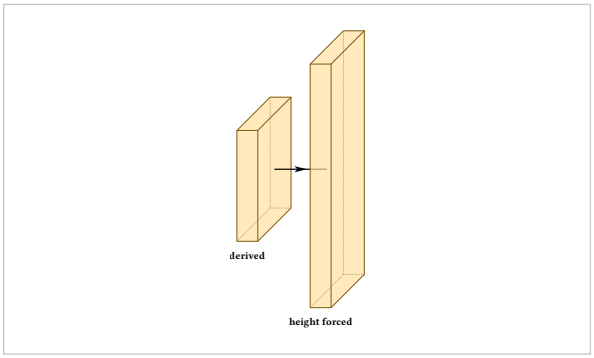
Both axes are logarithmic:

$$\text{height}, \text{depth} = 1.2 \log_2(\text{spatial}) - 3.2, \quad \text{width} = 0.075 \log_2(\text{channels})$$

Linear spatial extent does not work. Across a 640-to-20 pyramid it puts the smallest block at a quarter of a unit against 8 for the input, which is invisible.

Shape supplies defaults, not values
Anything you state explicitly wins, field by field, so a block can take its width and depth from its shape while its height is forced:

```
#draw-network((
  conv(label: "derived", shape: (256, 40, 40)),
  conv(
    label: "height forced",
    shape: (256, 40, 40),
    height: 7,
  ),
))
```



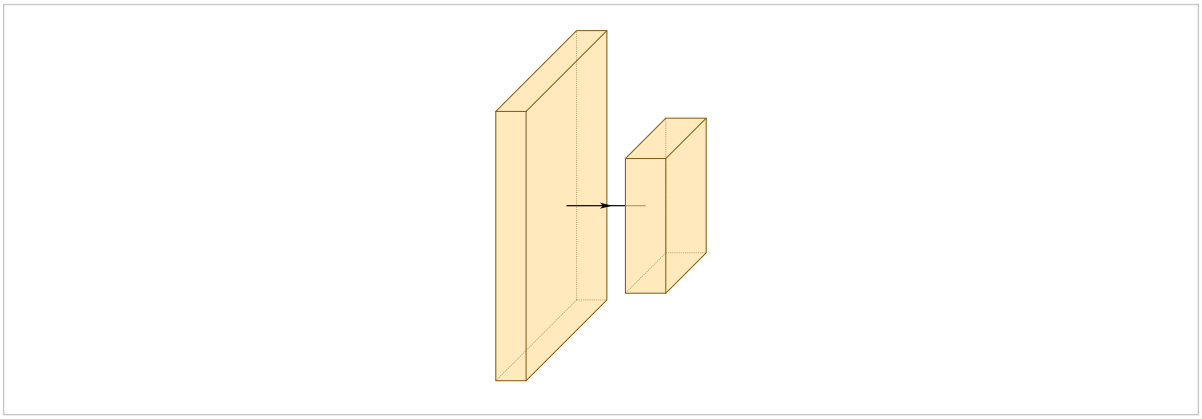
Changing the mapping

`shape-scale` default `(spatial: (1.2, -3.2), channels: (0.075, 0.0))`

`draw-network`

Slope and intercept for each axis, applied to the base-2 logarithm. The constants are absolute rather than normalised across a figure, so the same shape gives the same size everywhere and two figures stay comparable.

```
#draw-network(
  (conv(shape: (64, 128, 128)), conv(shape: (256, 32, 32))),
  shape-scale: (
    spatial: (2.0, -6.0),
    channels: (0.15, 0.0),
  ),
)
```

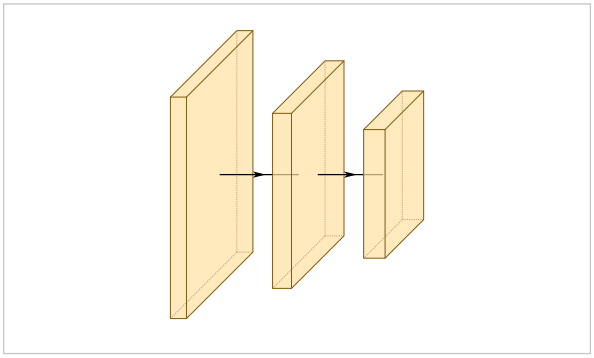


Only `conv`, `convres` and `custom` take a derived width, because only those three use thickness to mean channel count. The others have a fixed thickness that says something else, and keep it.

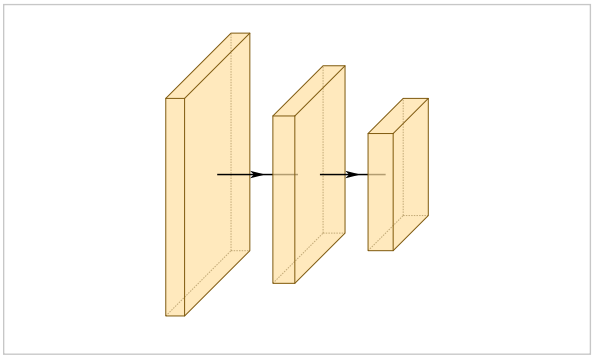
4.4 By hand against by shape

The same three blocks, written both ways. The first drifts the moment the model changes; the second cannot.

```
#draw-network((
  conv(height: 6.2, depth: 6.2, widths: (0.45,)),
  conv(height: 4.9, depth: 4.9, widths: (0.53,)),
  conv(height: 3.6, depth: 3.6, widths: (0.60,)),
))
```



```
#draw-network((
  conv(shape: (64, 128, 128)),
  conv(shape: (128, 64, 64)),
  conv(shape: (256, 32, 32)),
))
```



CHAPTER 5

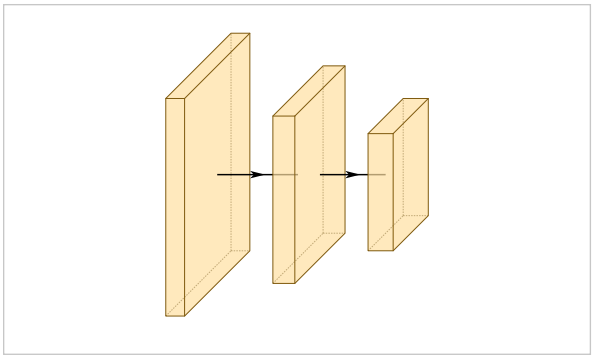
Spacing

`offset`

default `auto`

The gap between this layer and the one before it. `auto` computes it from the drawing; a number states it directly.

```
#draw-network((
  conv(shape: (64, 128, 128)),
  conv(shape: (128, 64, 64)),
  conv(shape: (256, 32, 32)),
))
```

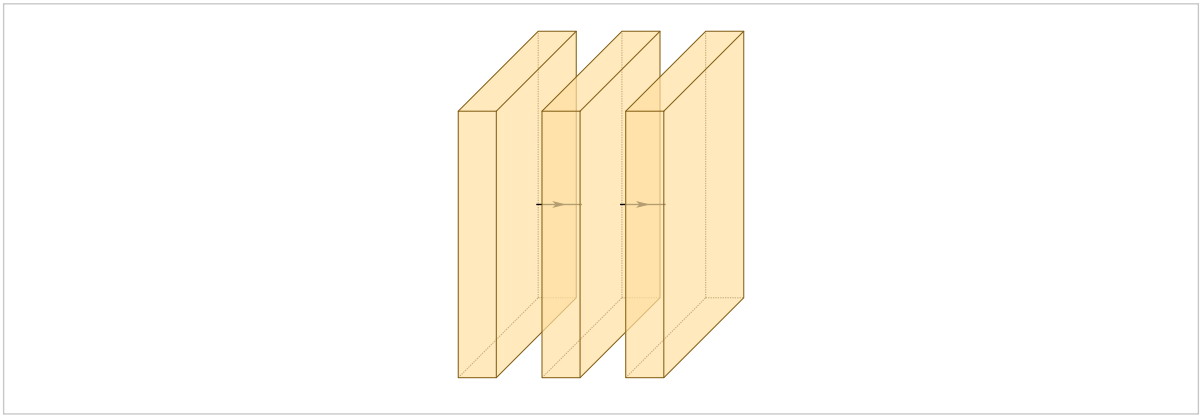


Automatic spacing covers the previous block's isometric lean, adds a fixed strip of white space, and widens where a connection descends into the gap.

5.1 Why a constant offset fails

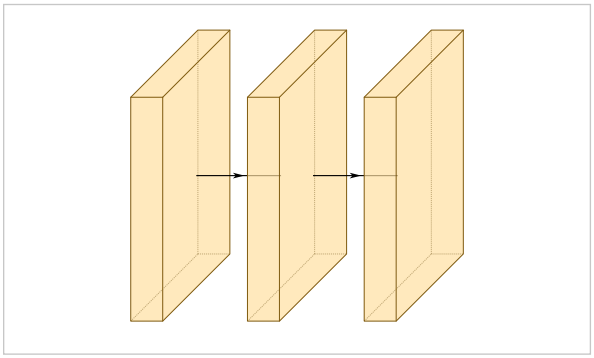
A block drawn in isometric projection leans right by `depth × depth-multiplier`. A gap narrower than that lean buys no visible space at all: the next block is drawn inside the previous one's sheared face, and the arrow between them disappears behind it.

```
#draw-network((
  conv(height: 7, depth: 7),
  conv(height: 7, depth: 7, offset: 1.2),
  conv(height: 7, depth: 7, offset: 1.2),
))
```



`offset: 1.2` against a depth-7 block, whose lean is 2.1.

```
#draw-network((
  conv(height: 7, depth: 7),
  conv(height: 7, depth: 7),
  conv(height: 7, depth: 7),
))
```



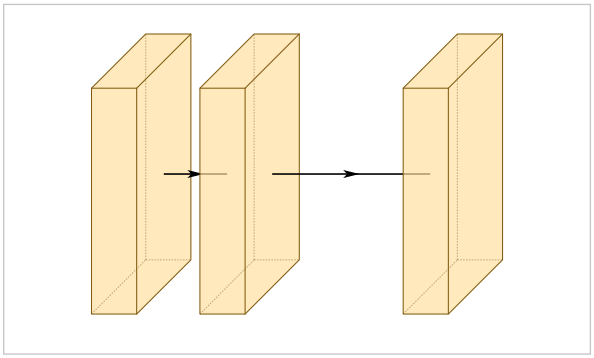
The same blocks with the offsets removed.

That is why `auto` is the default. A constant that works for one size of block fails for the next, and a figure that computes its own offsets ends up restating the size pyramid twice, with nothing to catch the two copies drifting apart.

5.2 Spacing by hand

A number overrides the automatic gap for one layer, and means what it always meant:

```
#draw-network((
  conv(depth: 4),
  conv(depth: 4, offset: 1.4),
  conv(depth: 4, offset: 3.5),
))
```

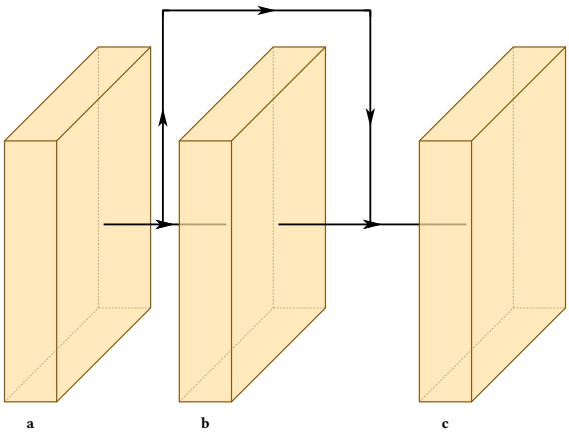


Two helpers are exported for figures that do compute their own geometry:

```
depth-shear(6)
min-clear-offset(6)
```

A connection descending between two blocks arrives at the midpoint of the arrow joining them, so it clears the lean only when half the offset exceeds it. That is what `min-clear-offset` returns, and what automatic spacing applies for you:

```
#draw-network(
  (
    conv(name: "a", label: "a", depth: 6),
    conv(name: "b", label: "b", depth: 6),
    conv(name: "c", label: "c", depth: 6),
  ),
  connections: (connection(from: "a", to: "c"),),
)
```



Both helpers take a `depth-multiplier` argument, which must match the one passed to `draw-network`. The default multiplier of `0.3` has no exact binary representation, so `depth-shear(4.5)` is `1.3499999999999999`. Compare with a tolerance, if you compare at all.

5.3 The white space constant

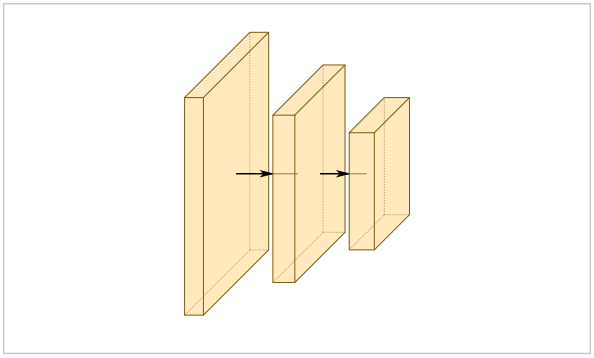
`auto-gap`

default `0.55`

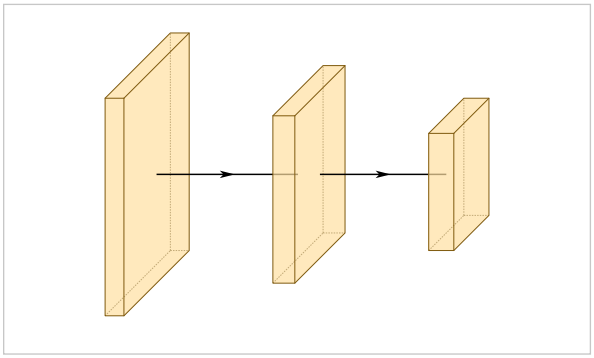
`draw-network`

How much visible white space an automatic offset adds, once the lean is covered.

```
#draw-network(
  (
    conv(shape: (64, 128, 128)),
    conv(shape: (128, 64, 64)),
    conv(shape: (256, 32, 32)),
  ),
  auto-gap: 0.1,
)
```



```
#draw-network(
  (
    conv(shape: (64, 128, 128)),
    conv(shape: (128, 64, 64)),
    conv(shape: (256, 32, 32)),
  ),
  auto-gap: 2.0,
)
```



CHAPTER 6

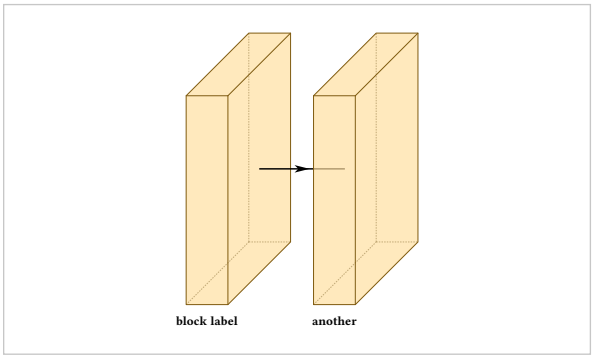
Labels

`label`

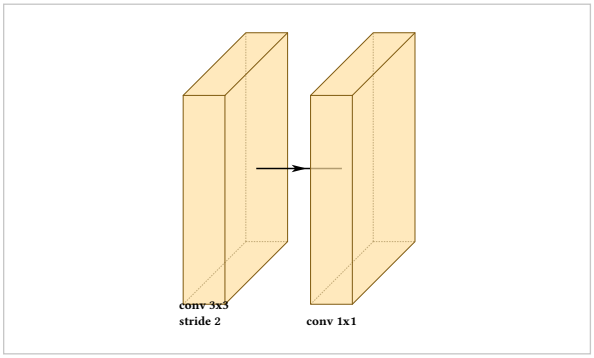
default none

Text under the block. Any content, so markup and line breaks work.

```
#draw-network((
  conv(label: "block label"),
  conv(label: "another"),
))
```




```
#draw-network((
  conv(label: [conv 3x3 \ stride 2]),
  conv(label: [conv 1x1]),
))
```



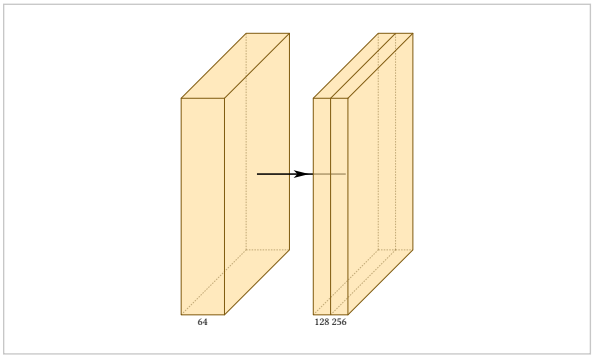
6.1 Channel numbers

`channels`

default none

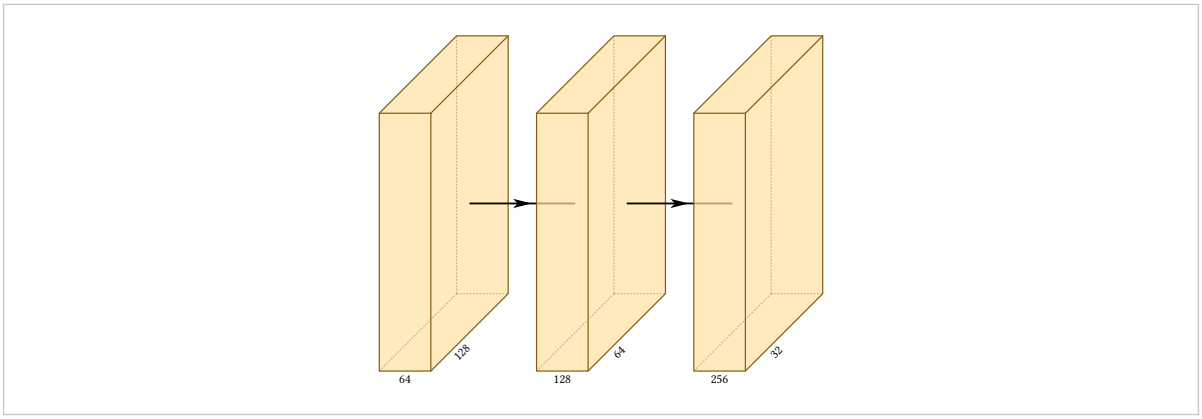
Numbers along the top of the block, one per band. At most one extra entry is allowed, and becomes the diagonal axis label. Strings work as well as numbers.

```
#draw-network((
  conv(channels: (64,)),
  conv(channels: (128, 256), widths: (0.4, 0.4)),
))
```



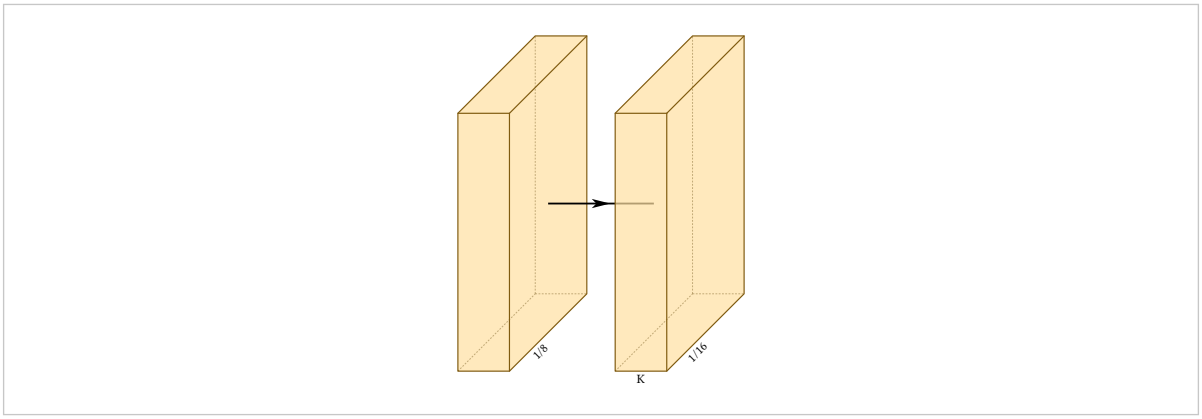
The extra entry is where spatial resolution usually goes:

```
#draw-network((
  conv(channels: (64, 128)),
  conv(channels: (128, 64)),
  conv(channels: (256, 32)),
))
```



An empty string labels the axis and nothing else:

```
#draw-network((
  conv(channels: ("", "1/8")),
  conv(channels: ("K", "1/16")),
))
```



More than one extra entry is an error from inside the package, reported as an array index out of bounds rather than as a message about `channels`. If you see that, count your bands.

6.2 Axis names

`xlabel` default none

conv, convres, custom

`ylabel` default none

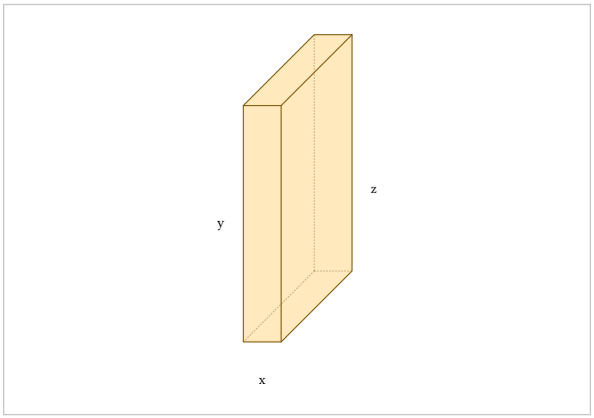
conv, convres, custom

`zlabel` default none

conv, convres, custom

Names for the three axes of one block, for the figure that has to explain the projection itself.

```
#draw-network((
  conv(
    xlabel: "x", ylabel: "y", zlabel: "z",
    widths: (0.8,),
  ),
))
```

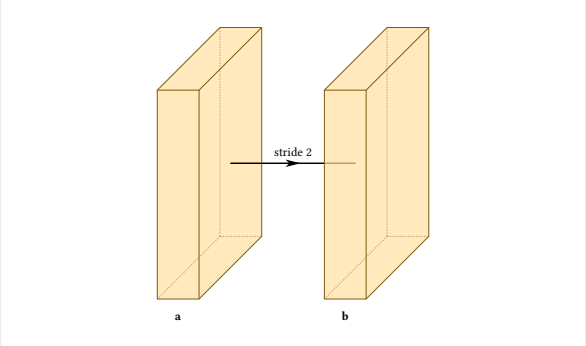


6.3 Labelling the arrow between two blocks

`connection-label` default none

Text on the axis arrow **after** the layer that declares it.

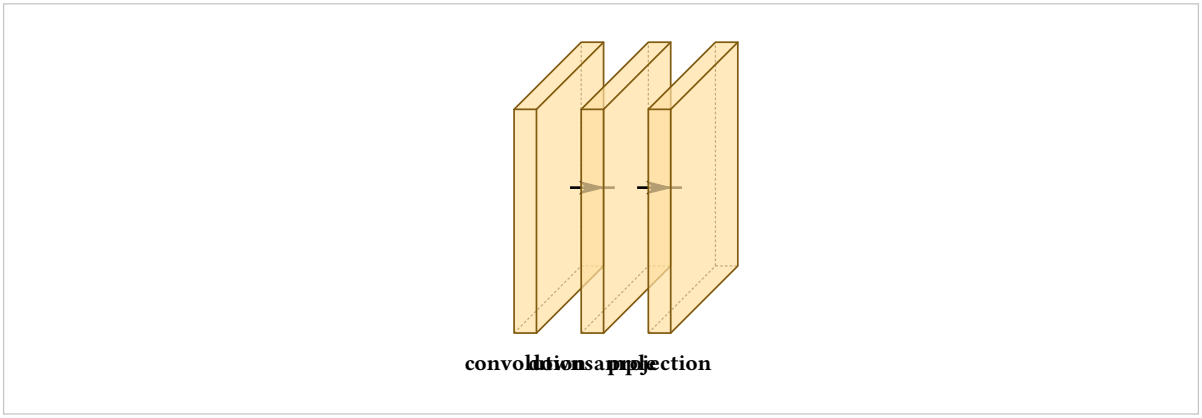
```
#draw-network((
  conv(label: "a", connection-label: "stride 2"),
  conv(label: "b", offset: 3),
))
```



6.4 When labels collide

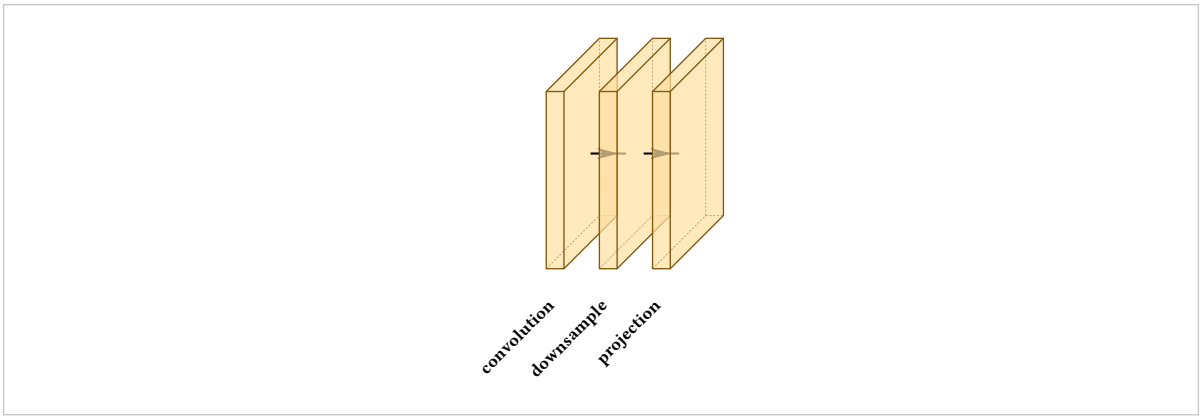
Labels sit at a fixed spot beneath each block, so long labels on closely spaced blocks overlap:

```
#let b = (height: 3, depth: 3, widths: (0.3,), offset: 0.6)
#draw-network((
  conv(label: "convolution", ..b),
  conv(label: "downsample", ..b),
  conv(label: "projection", ..b),
))
```



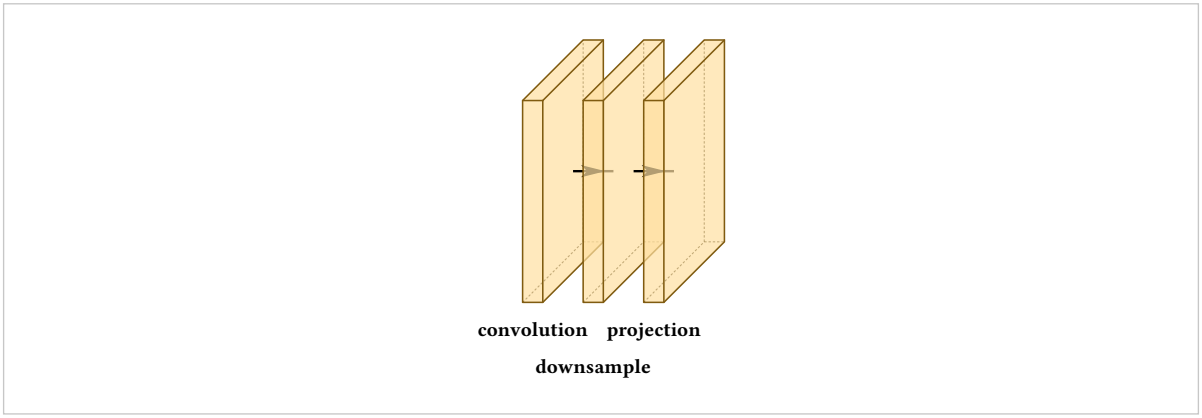
There are three fixes, and which to reach for depends on how crowded the row is. Rotating trades horizontal space, which is scarce, for vertical space, which is usually free:

```
#let b = (
  height: 3, depth: 3, widths: (0.3,), offset: 0.6,
  label-orient: "diagonal",
)
#draw-network((
  conv(label: "convolution", ..b),
  conv(label: "downsample", ..b),
  conv(label: "projection", ..b),
))
```



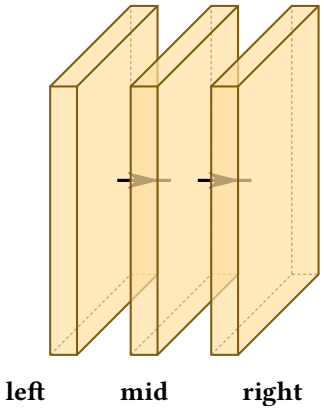
Dropping one label to a second row clears a single collision without rotating anything:

```
#let b = (height: 3, depth: 3, widths: (0.3,), offset: 0.6)
#draw-network((
  conv(label: "convolution", ..b),
  conv(label: "downsample", label-dy: -0.55, ..b),
  conv(label: "projection", ..b),
))
```



Anchoring the outer labels by their facing edges fans them sideways:

```
#let b = (height: 3, depth: 3, widths: (0.3,), offset: 0.6)
#draw-network((
  conv(label: "left", label-anchor: "base-east",
       label-dx: -0.2, ..b),
  conv(label: "mid", ..b),
  conv(label: "right", label-anchor: "base-west",
       label-dx: 0.2, ..b),
))
```



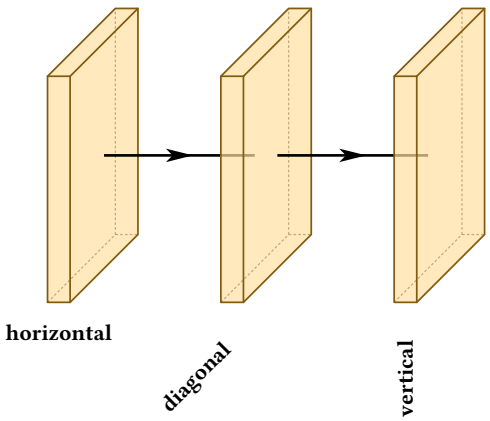
6.5 The placement options

`label-orient`

default `"horizontal"`

`"horizontal"`, `"diagonal"` or `"vertical"`. Each preset pairs its angle with the anchor that places it correctly, so a diagonal label points its end at its own block and a vertical one hangs centred beneath it. Any other value is an error.

```
#let b = (height: 3, depth: 3, widths: (0.3,), offset: 2)
#draw-network((
  conv(label: "horizontal", ..b),
  conv(label: "diagonal", label-orient: "diagonal", ..b),
  conv(label: "vertical", label-orient: "vertical", ..b),
))
```



`label-dx`

default `0`

Horizontal shift.

`label-dy`

default `0`

Vertical shift.

`label-anchor`

default `set by the preset`

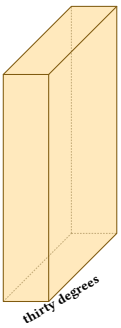
Which point of the text is pinned to the label position. Labels anchor on their baseline rather than their bounding box, so a label with descenders sits level with one without. Use `"base-east"` and `"base-west"` to anchor horizontally while keeping that alignment.

`label-angle`

default `0deg`

An arbitrary angle. Setting both this and `label-orient` is an error, since the right anchor for an arbitrary rotation depends on where you want the text.

```
#draw-network((
  conv(
    label: "thirty degrees",
    label-angle: 30deg,
    label-anchor: "base-west",
  ),
))
```



CHAPTER 7

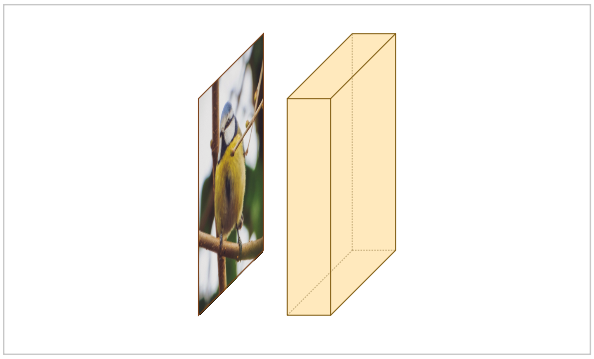
Images inside blocks

`image`

default none

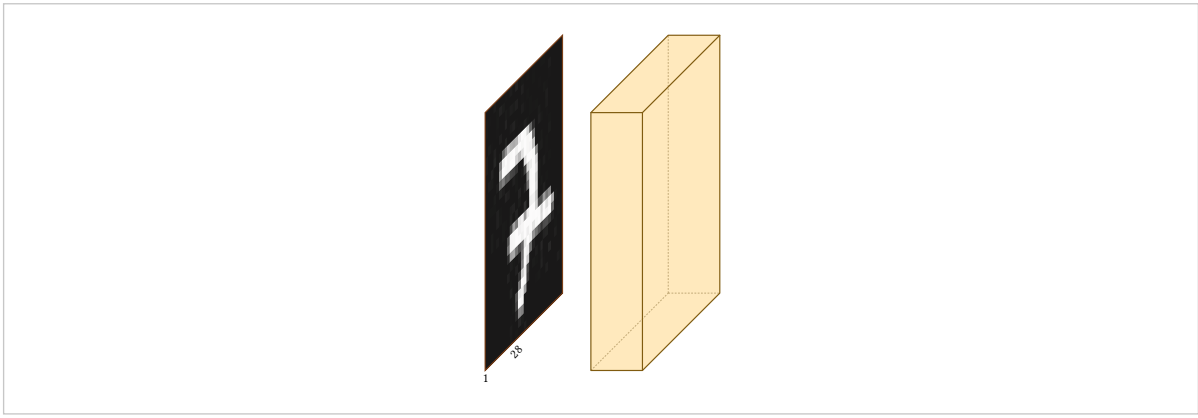
Content drawn on the block's face, sheared to match the projection. The string `"default"` uses the bundled photograph. Any content works, not only images.

```
#draw-network((
  input(image: "default"),
  conv(),
))
```



Any Typst image:

```
#let mnist = "../../examples/networks/mnist-img-sample.jpg"
#draw-network((
  input(image: image(mnist), channels: (1, 28)),
  conv(),
))
```

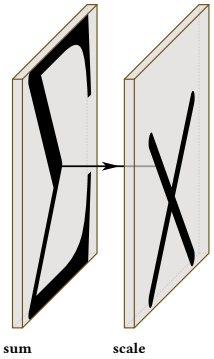


Nothing restricts this to the input. Putting a picture on a later block is how a figure shows what a stage produces:

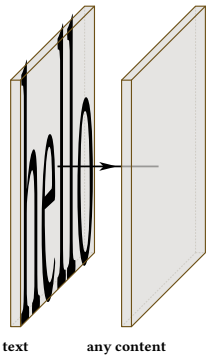
```
#draw-network((
  input(image: "default", label: "in"),
  conv(image: "default", label: "features",
       depth: 4, widths: (0.4,), offset: 2),
))
```

7.1 Content that is not an image

```
#draw-network((
  custom(image: [#text(20pt)[$Sigma$]], label: "sum"),
  custom(image: [#text(20pt)[$times$]], label: "scale"),
))
```



```
#draw-network((
  custom(image: [hello], label: "text"),
  custom(image: [#rect(width: 60%, height: 60%, fill: red)],
         label: "any content"),
))
```



CHAPTER 8

Colour

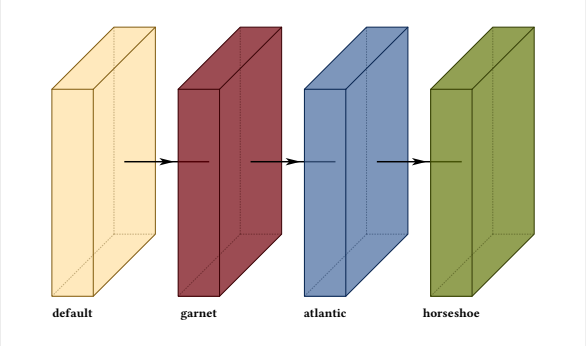
`fill` default `set by the type and palette`

Block colour. Edge and band colours are derived from it, so a recoloured block stays internally consistent.

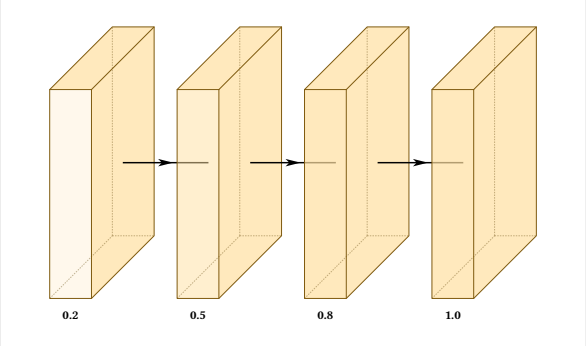
`opacity` default `0.7, varies by type`

How solid the block is.

```
#draw-network((
  conv(label: "default"),
  conv(label: "garnet", fill: rgb("#73000A")),
  conv(label: "atlantic", fill: rgb("#466A9F")),
  conv(label: "horseshoe", fill: rgb("#65780B")),
))
```




```
#draw-network((
  conv(label: "0.2", opacity: 0.2),
  conv(label: "0.5", opacity: 0.5),
  conv(label: "0.8", opacity: 0.8),
  conv(label: "1.0", opacity: 1.0),
))
```



8.1 Activation bands

A darker band at the end of a convolution conventionally marks the activation.

`show-relu` default `false`

`draw-network`; `conv`, `convres`, `custom`

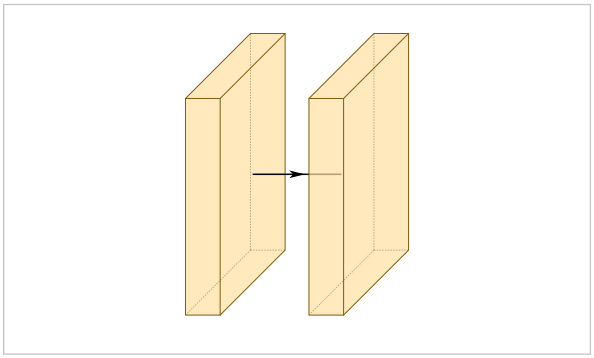
Turn bands on for a whole figure with the `draw-network` argument, and override on any one layer.

`bandfill` default `derived from fill`

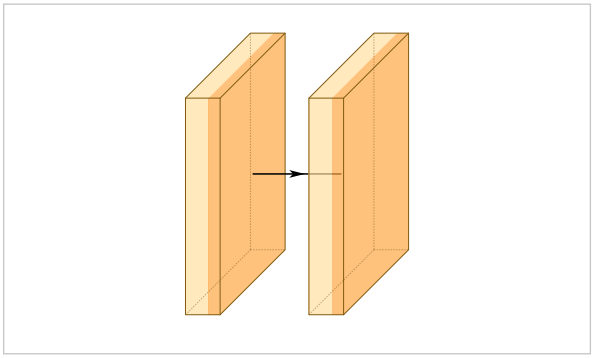
`conv`, `convres`, `custom`

Band colour. Undeclared, it is derived from the layer's own fill, so a recoloured block gets a matching band rather than one from the palette.

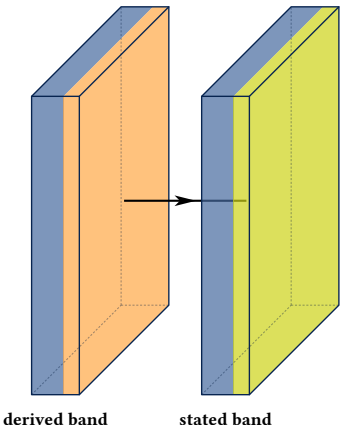
```
#draw-network(
  (conv(widths: (0.8,)), conv(widths: (0.8,))),
  show-relu: false,
)
```



```
#draw-network(
  (conv(widths: (0.8,)), conv(widths: (0.8,))),
  show-relu: true,
)
```



```
#draw-network(
  (
    conv(label: "derived band", widths: (0.8,),
         fill: rgb("#466A9F")),
    conv(label: "stated band", widths: (0.8,),
         fill: rgb("#466A9F"), bandfill: rgb("#CED318")),
  ),
  show-relu: true,
)
```



8.2 Palettes

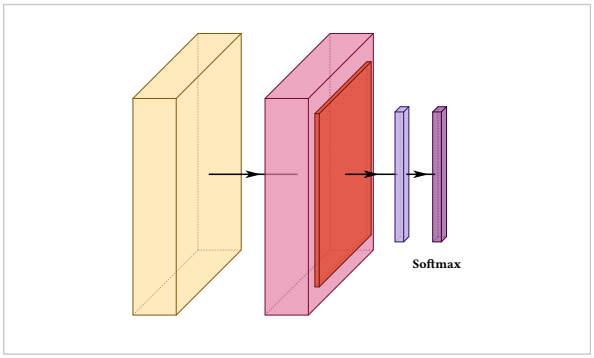
palette

default `"warm"`

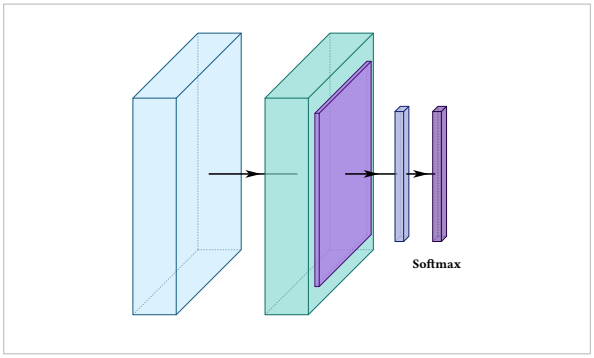
draw-network

`"warm"` or `"cold"`.

```
#draw-network(
  (conv(), convres(), pool(), fc(), softmax()),
  palette: "warm",
)
```

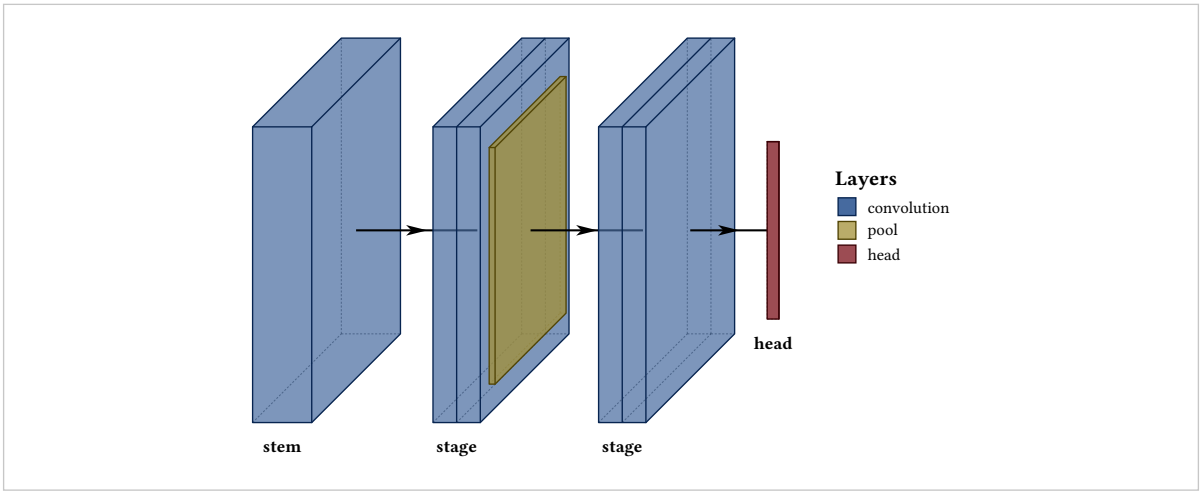



```
#draw-network(
  (conv(), convres(), pool(), fc(), softmax()),
  palette: "cold",
)
```



A palette sets defaults only. Any `fill` you state wins, so a figure can sit on a palette and still recolour three blocks — or ignore the palette entirely and put the whole figure on a scheme of your own:

```
#let garnet = rgb("#73000A")
#let atlantic = rgb("#466A9F")
#let honeycomb = rgb("#A49137")

#draw-network(
  (
    conv(label: "stem", fill: atlantic, legend: "convolution"),
    conv(label: "stage", fill: atlantic, widths: (0.4, 0.4)),
    pool(fill: honeycomb, legend: "pool"),
    conv(label: "stage", fill: atlantic, widths: (0.4, 0.4)),
    fc(label: "head", fill: garnet, depth: 0, legend: "head"),
  ),
  show-legend: true,
)
```

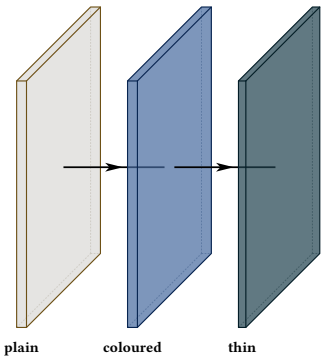


CHAPTER 9

Blocks of your own

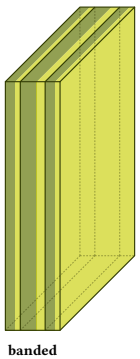
`custom` is the generic block: no meaning attached, every option available.

```
#draw-network((
  custom(label: "plain"),
  custom(label: "coloured", fill: rgb("#466A9F")),
  custom(label: "thin", width: 0.15, fill: rgb("#1F414D")),
))
```



It takes bands and an activation band exactly as a convolution does:

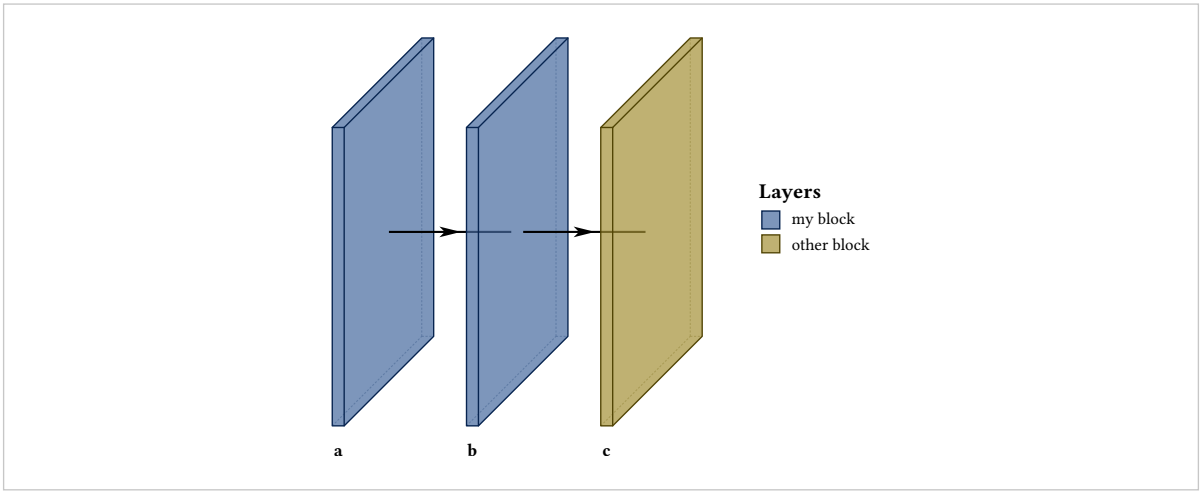
```
#draw-network(
  (
    custom(
      label: "banded",
      widths: (0.3, 0.5, 0.3),
      fill: rgb("#65780B"),
      bandfill: rgb("#CED318"),
      show-relu: true,
    ),
  ),
)
```



`legend` default `the type's default name`

What this layer is called in the legend. Layers sharing a legend name appear once, so a colour used by four blocks produces one entry.

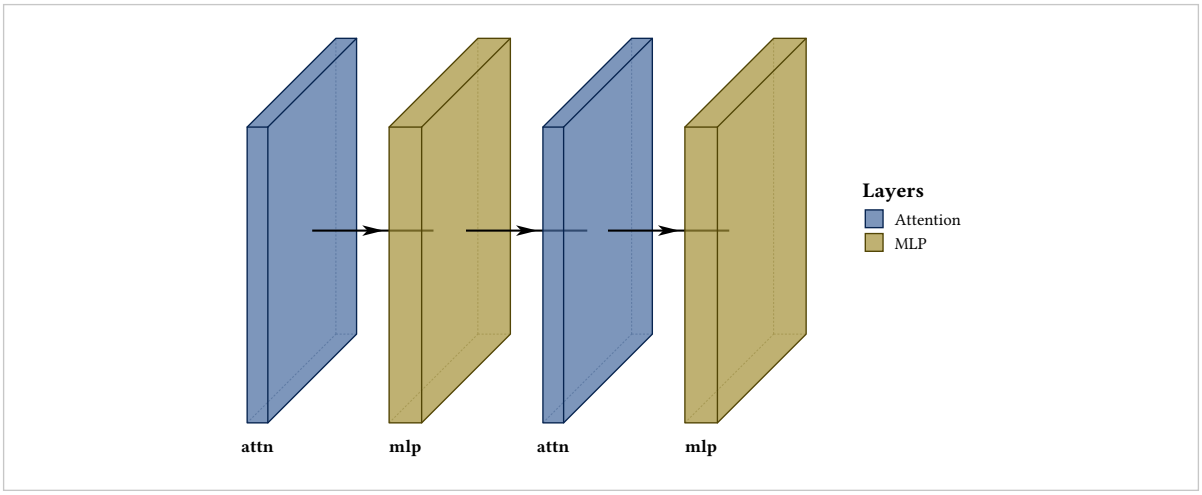
```
#draw-network(
  (
    custom(label: "a", fill: rgb("#466A9F"), legend: "my block"),
    custom(label: "b", fill: rgb("#466A9F")),
    custom(label: "c", fill: rgb("#A49137"), legend: "other block"),
  ),
  show-legend: true,
)
```



9.1 Building a vocabulary

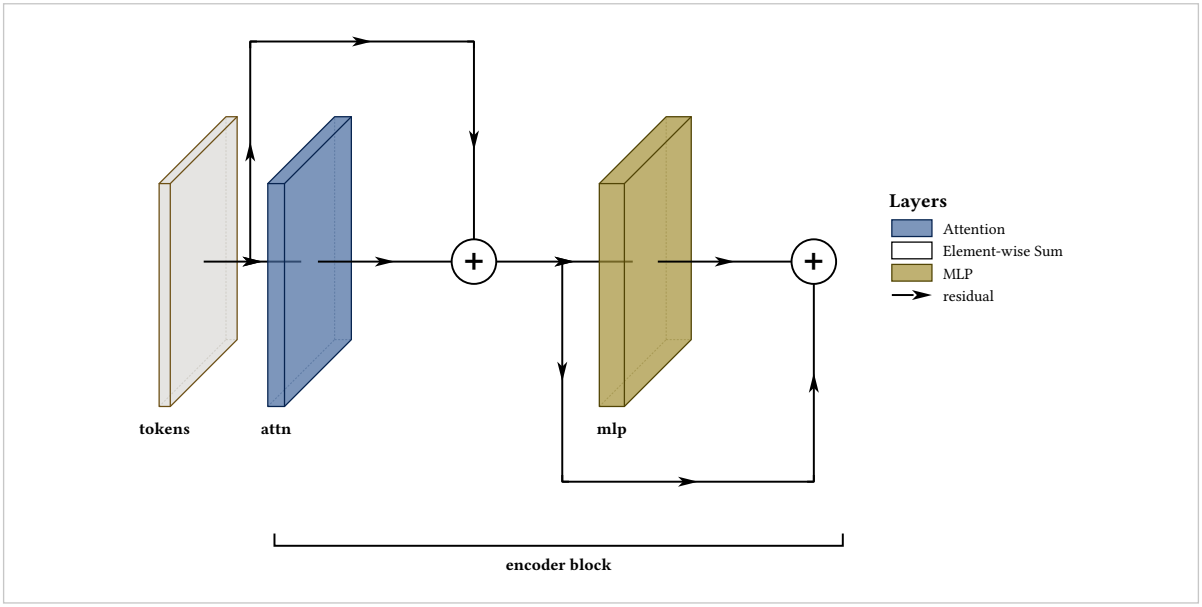
A `custom` layer with its options fixed is a function returning a layer, so your own block types are a few lines. This is the intended way to draw something the package has no name for — attention, a gate, a state-space block — rather than repurposing a type that means something else to a reader.

```
#let attention(..a) = custom(
  fill: rgb("#466A9F"), width: 0.35, legend: "Attention", ..a,
)
#let mlp(..a) = custom(
  fill: rgb("#A49137"), width: 0.55, legend: "MLP", ..a,
)

#draw-network(
  (attention(label: "attn"), mlp(label: "mlp"),
   attention(label: "attn"), mlp(label: "mlp")),
  show-legend: true,
)
```



Once the vocabulary exists, a figure is written in it:

```
#let attn(..a) = custom(fill: rgb("#466A9F"), width: 0.3,
  height: 4, depth: 4, legend: "Attention", ..a)
#let mlp(..a) = custom(fill: rgb("#A49137"), width: 0.45,
  height: 4, depth: 4, legend: "MLP", ..a)

#draw-network(
  (
    custom(name: "in", label: "tokens", width: 0.2, height: 4, depth: 4),
    attn(name: "a", label: "attn"),
    sum(name: "s1"),
    mlp(name: "m", label: "mlp"),
    sum(name: "s2"),
  ),
  connections: (
    connection(from: "in", to: "s1", legend: "residual"),
    connection(from: "s1", to: "s2"),
  ),
  groups: (group(from: "a", to: "s2", label: "encoder block"),),
  show-legend: true,
)
```



CHAPTER 10

Connections

Arrows along the main axis are drawn for you. Anything else — a skip, a residual add, a lateral feed, an auxiliary path — goes in `connections`, as a list of routes between two named layers.

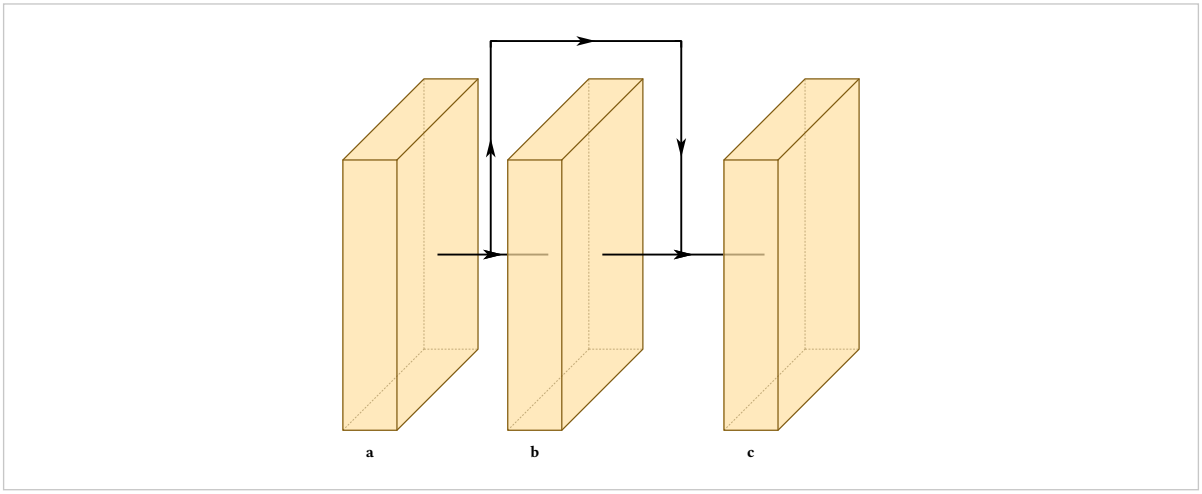
`from` default `required`

Name of the layer the route leaves.

`to` default `required`

Name of the layer it arrives at.

```
#draw-network(
  (
    conv(name: "a", label: "a"),
    conv(name: "b", label: "b"),
    conv(name: "c", label: "c"),
  ),
  connections: (connection(from: "a", to: "c"),),
)
```

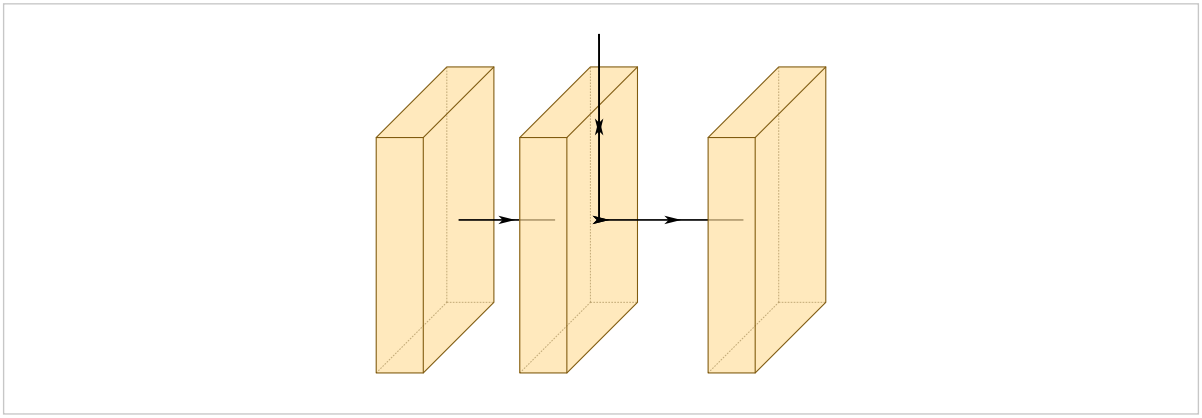


10.1 Routing modes

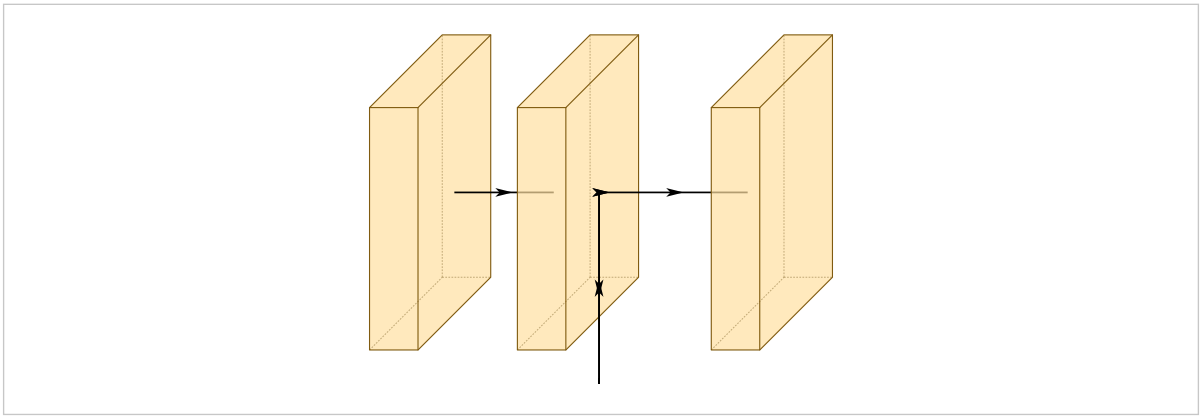
`mode` default `"air"`

`"air"` runs over the top of the stack, `"flat"` underneath, `"depth"` along the projection's own axis.

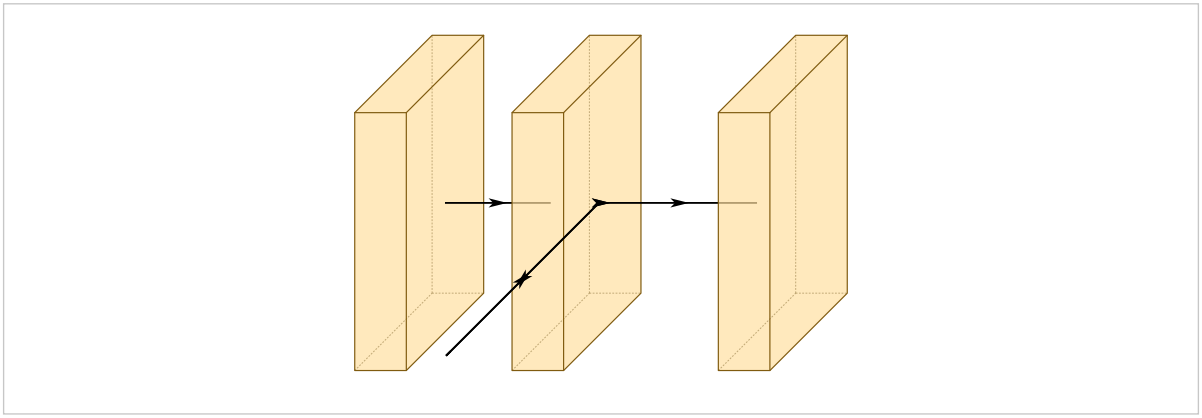
```
#draw-network(
  (conv(name: "a"), conv(), conv(name: "c")),
  connections: (connection(from: "a", to: "c", mode: "air"),),
)
```



```
#draw-network(
  (conv(name: "a"), conv(), conv(name: "c")),
  connections: (connection(from: "a", to: "c", mode: "flat"),),
)
```



```
#draw-network(
  (conv(name: "a"), conv(), conv(name: "c")),
  connections: (connection(from: "a", to: "c", mode: "depth"),),
)
```



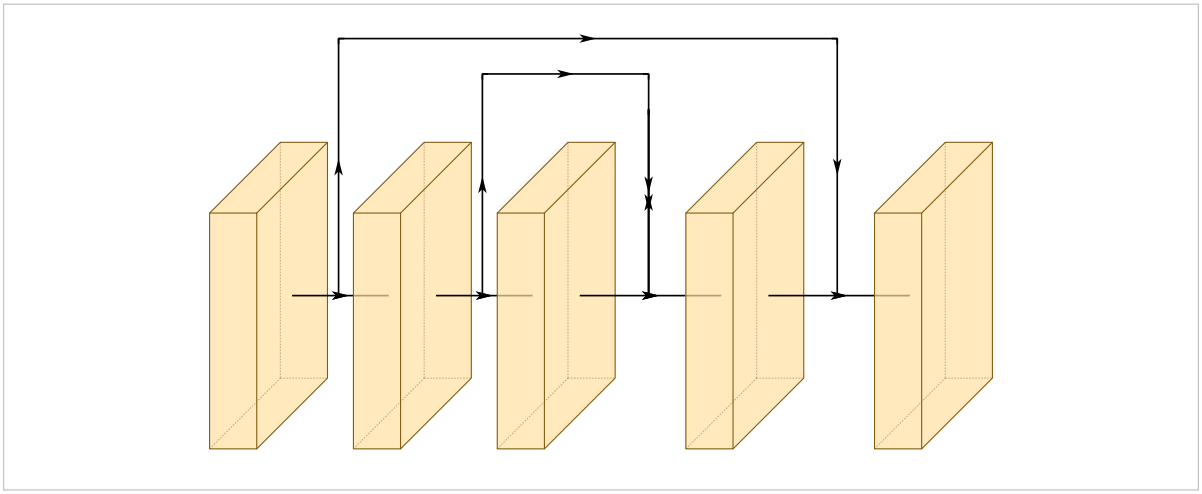
10.2 Lane heights

pos

default `auto`

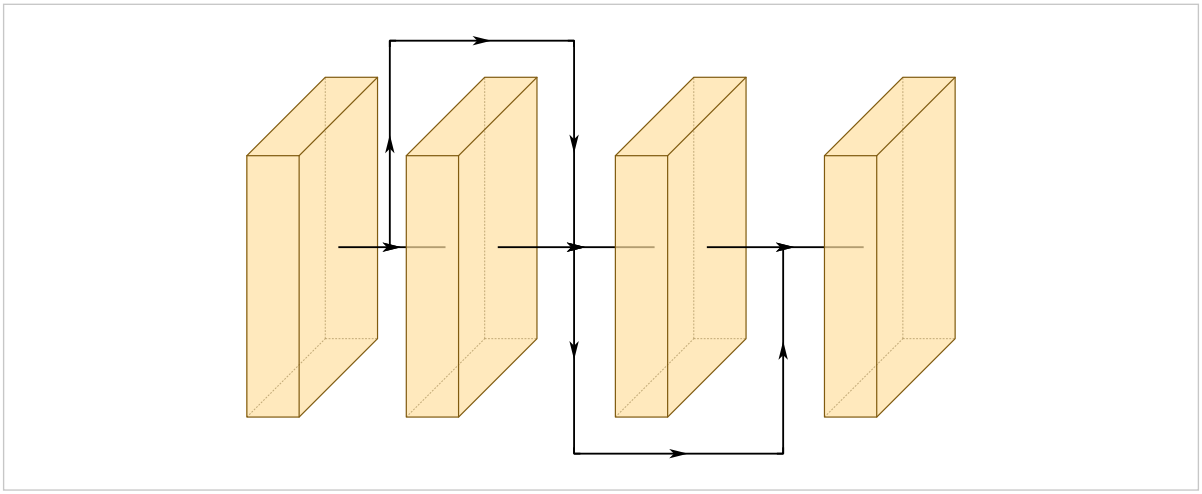
How far the route sits from the centre axis. `auto` ranks routes by how far they reach, so a route spanning more blocks sits higher and a longer route always arcs over a shorter one instead of crossing it.

```
#draw-network(
  (
    conv(name: "a"), conv(name: "b"), conv(name: "c"),
    conv(name: "d"), conv(name: "e"),
  ),
  connections: (
    connection(from: "a", to: "e"),
    connection(from: "b", to: "d"),
    connection(from: "c", to: "d"),
  ),
)
```



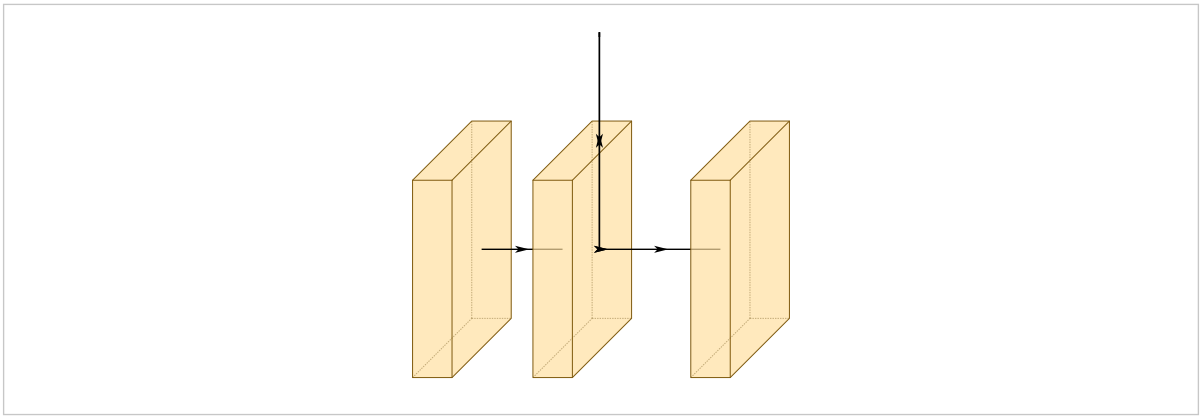
Routes of equal reach share a height. Where two of them overlap, the second goes to the opposite side of the axis at the same height, rather than being pushed further out than its reach warrants:

```
#draw-network(
  (
    conv(name: "a"), conv(name: "b"),
    conv(name: "c"), conv(name: "d"),
  ),
  connections: (
    connection(from: "a", to: "c"),
    connection(from: "b", to: "d"),
  ),
)
```



A number overrides all of it for one route:

```
#draw-network(
  (conv(name: "a"), conv(), conv(name: "c")),
  connections: (connection(from: "a", to: "c", pos: 5.5),),
)
```



`lane-unit` default `0.75`

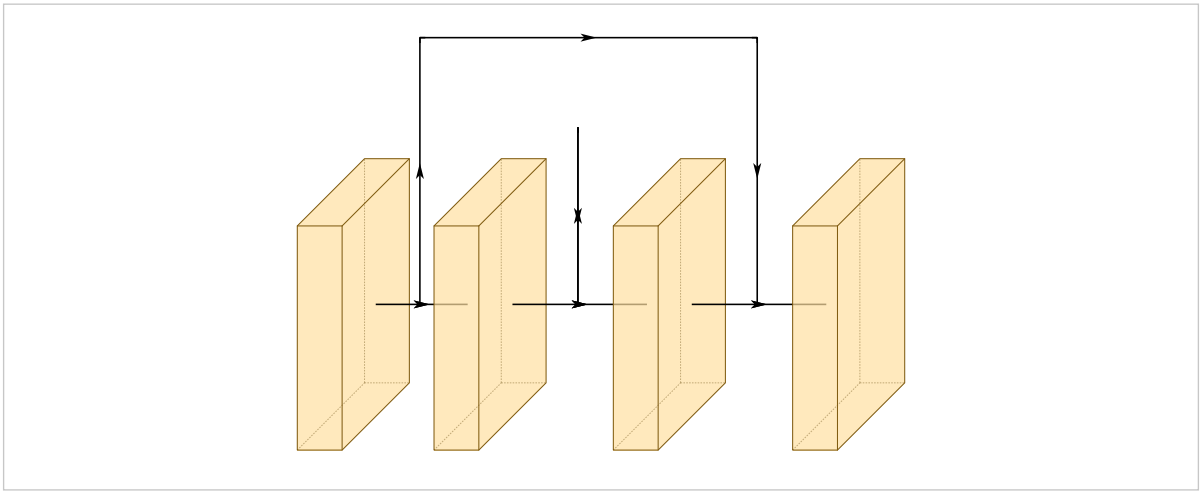
draw-network

Spacing between automatic heights.

`clearance` default `0.7`

How far the lowest automatic route sits from the blocks.

```
#draw-network(
  (
    conv(name: "a"), conv(name: "b"),
    conv(name: "c"), conv(name: "d"),
  ),
  connections: (
    connection(from: "a", to: "d"),
    connection(from: "b", to: "c"),
  ),
  lane-unit: 2.0,
)
```

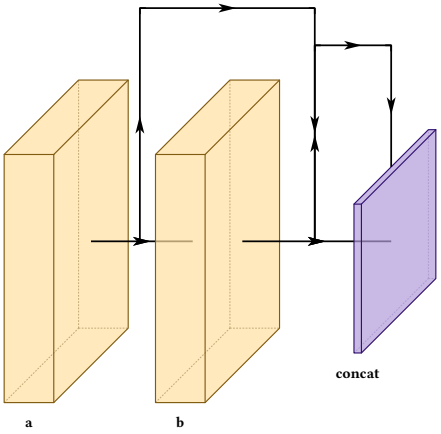


10.3 Where a route arrives

`touch-layer` default `false`

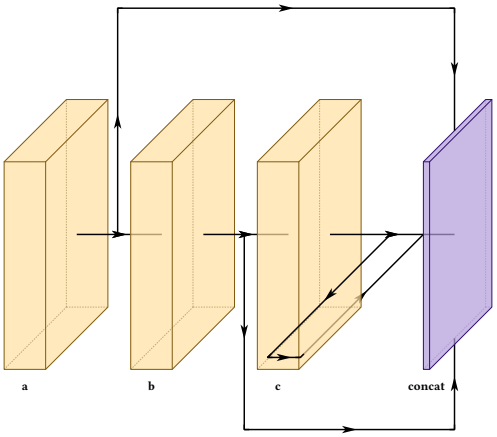
Arrive on the target block itself rather than on the axis arrow in front of it. The edge follows the routing mode: air lands on the top edge, flat on the bottom, depth on the left.

```
#draw-network(
  (
    conv(name: "a", label: "a"),
    conv(name: "b", label: "b"),
    concat(name: "cat", label: "concat", depth: 5),
  ),
  connections: (
    connection(from: "a", to: "cat"),
    connection(from: "b", to: "cat", touch-layer: true),
  ),
)
```



All three modes, arriving on the same block:

```
#let t = (touch-layer: true)
#draw-network(
  (
    conv(name: "a", label: "a"),
    conv(name: "b", label: "b"),
    conv(name: "c", label: "c"),
    concat(name: "cat", label: "concat", depth: 5, height: 5),
  ),
  connections: (
    connection(from: "a", to: "cat", mode: "air", ..t),
    connection(from: "b", to: "cat", mode: "flat", ..t),
    connection(from: "c", to: "cat", mode: "depth", ..t),
  ),
)
```



A route arriving on the bottom or left edge has its final stretch drawn behind the block, because those edges are on the far side and the route genuinely passes underneath before reaching them. Blocks are semi-transparent, so it shows faintly rather than disappearing.

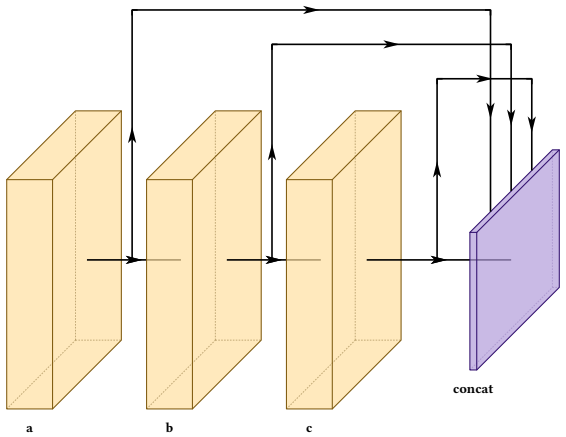
10.4 Fanning several routes into one block

Two routes in the same mode would otherwise land on the same point, with their arrowheads stacked.

`arrive-offset` default `auto`

Position along the arrival edge. `auto` spaces a whole fan evenly and leaves a lone arrival centred.

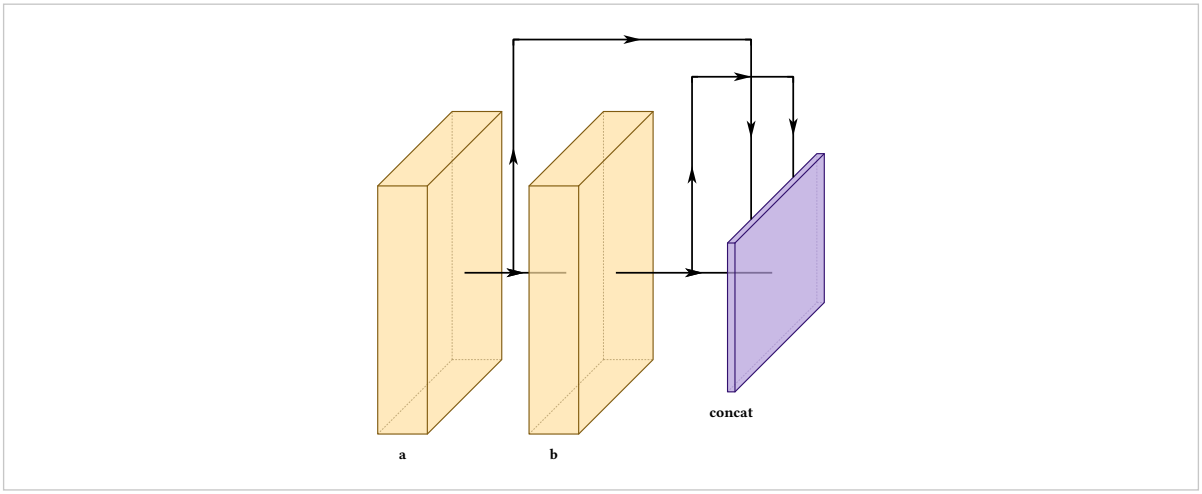
```
#draw-network(
  (
    conv(name: "a", label: "a"), conv(name: "b", label: "b"),
    conv(name: "c", label: "c"),
    concat(name: "cat", label: "concat", depth: 6),
  ),
  connections: (
    connection(from: "a", to: "cat", touch-layer: true),
    connection(from: "b", to: "cat", touch-layer: true),
    connection(from: "c", to: "cat", touch-layer: true),
  ),
)
```



Routes are grouped by the edge they land on, then spread across it: `k` routes divide the edge into `k + 1` intervals and sit at the interior boundaries, so the outermost pair is inset rather than sitting on the corners. They are ordered by where each route starts, so a fan does not cross itself, and adding a route respaces the rest.

A number places one arrival yourself, measured along the edge:

```
#draw-network(
  (
    conv(name: "a", label: "a"), conv(name: "b", label: "b"),
    concat(name: "cat", label: "concat", depth: 6),
  ),
  connections: (
    connection(from: "a", to: "cat", touch-layer: true, arrive-offset: -0.6),
    connection(from: "b", to: "cat", touch-layer: true, arrive-offset: 0.6),
  ),
)
```



The offset runs **along the edge**, not in x. The top and bottom edges of a block's west side follow the isometric depth direction, so shifting horizontally would walk the arrival off the block. The edge is $\sqrt{2} \cdot \text{depth} \cdot \text{depth-multiplier}$ for top and bottom arrivals and the block height for left ones, so a deeper block accommodates a wider fan.

10.5 Telling routes apart

A residual add, a concat feed and an auxiliary supervision path are different things, and by default they all look the same.

`color` default `palette`

Paint for the line and its arrowheads.

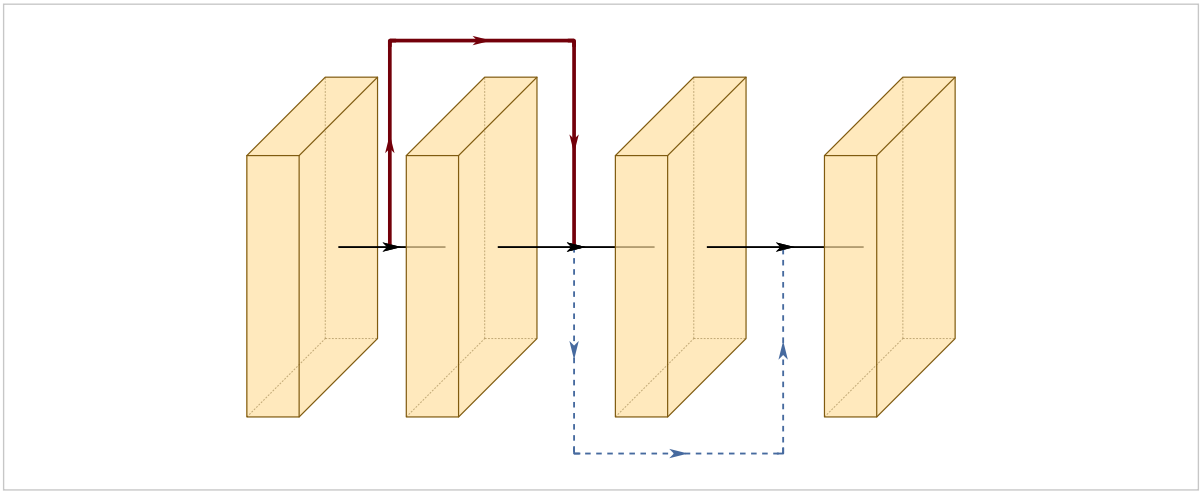
`dash` default `none`

Any Typst dash pattern, e.g. `"dashed"`, `"dotted"`.

`thickness` default `1`

Multiplies the palette stroke width, so it composes with `stroke-thickness` rather than fighting it.

```
#draw-network(
  (
    conv(name: "a"), conv(name: "b"),
    conv(name: "c"), conv(name: "d"),
  ),
  connections: (
    connection(from: "a", to: "c", color: rgb("#73000A"), thickness: 2),
    connection(from: "b", to: "d", color: rgb("#466A9F"), dash: "dashed"),
  ),
)
```



Arrowheads take the line colour, since a coloured line with black arrowheads reads as a bug rather than a choice.

10.6 Routes in the legend

`legend` default `none`

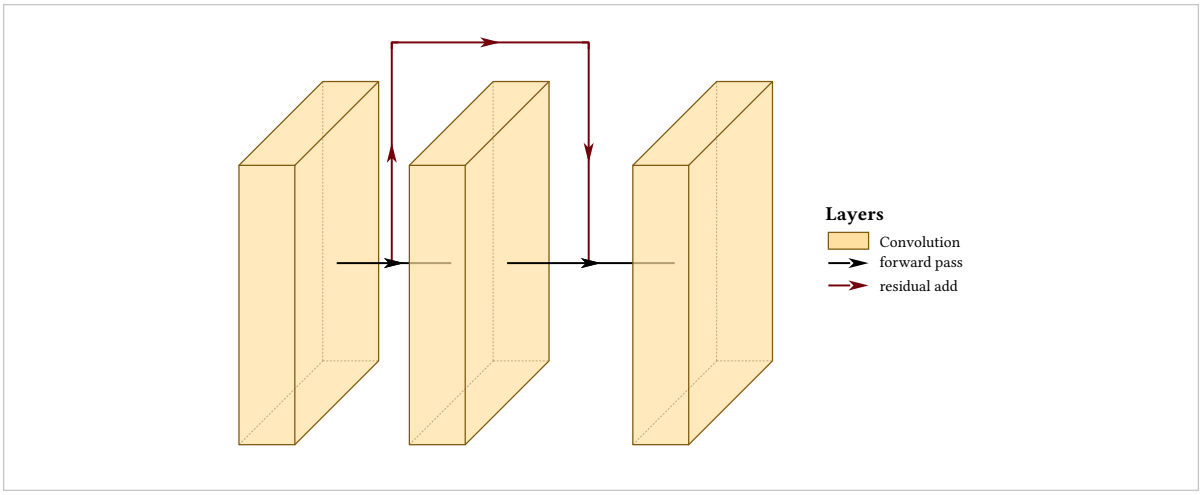
Adds a line-sample legend entry rather than a colour swatch, since what distinguishes a route is its stroke and not a fill. Routes sharing a name appear once.

`main-legend` default `none`

`draw-network`

Names the automatic axis arrows. Without it a legend can explain every skip in a figure and say nothing about the arrows carrying the forward pass.

```
#draw-network(
  (conv(name: "a"), conv(name: "b"), conv(name: "c")),
  connections: (
    connection(from: "a", to: "c", color: rgb("#73000A"),
     legend: "residual add"),
  ),
  show-legend: true,
  main-legend: "forward pass",
)
```



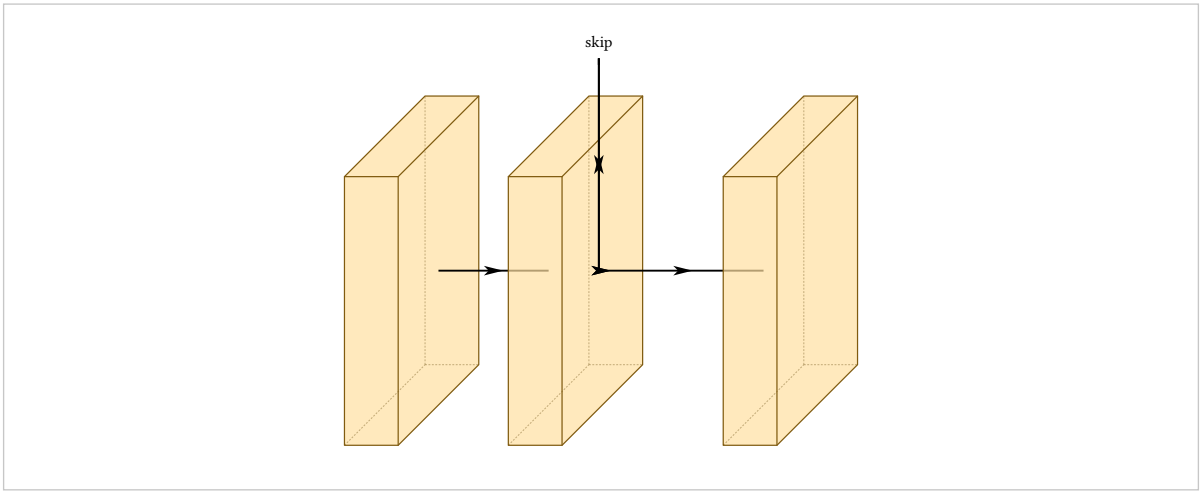
10.7 Labelling a route

`label`

default `none`

Text on the route.

```
#draw-network(
  (conv(name: "a"), conv(), conv(name: "c")),
  connections: (connection(from: "a", to: "c", label: "skip"),),
)
```



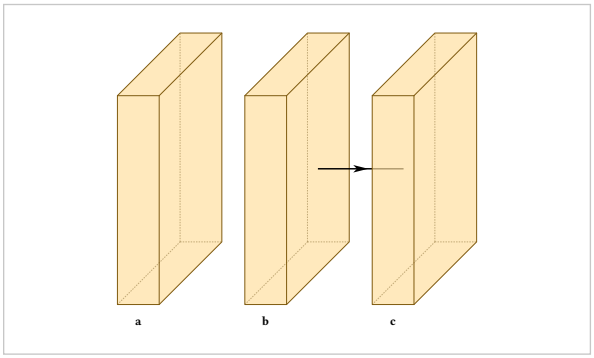
10.8 Suppressing an axis arrow

`show-connection`

default `true; false on input`

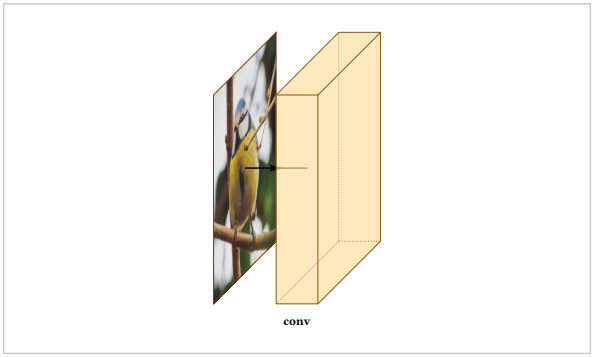
Whether to draw the axis arrow **after** this layer.

```
#draw-network((
  conv(label: "a", show-connection: false),
  conv(label: "b"),
  conv(label: "c"),
))
```



The input layer is the one that defaults to `false`, which is why an image usually sits detached. Set it `true` to join it up:

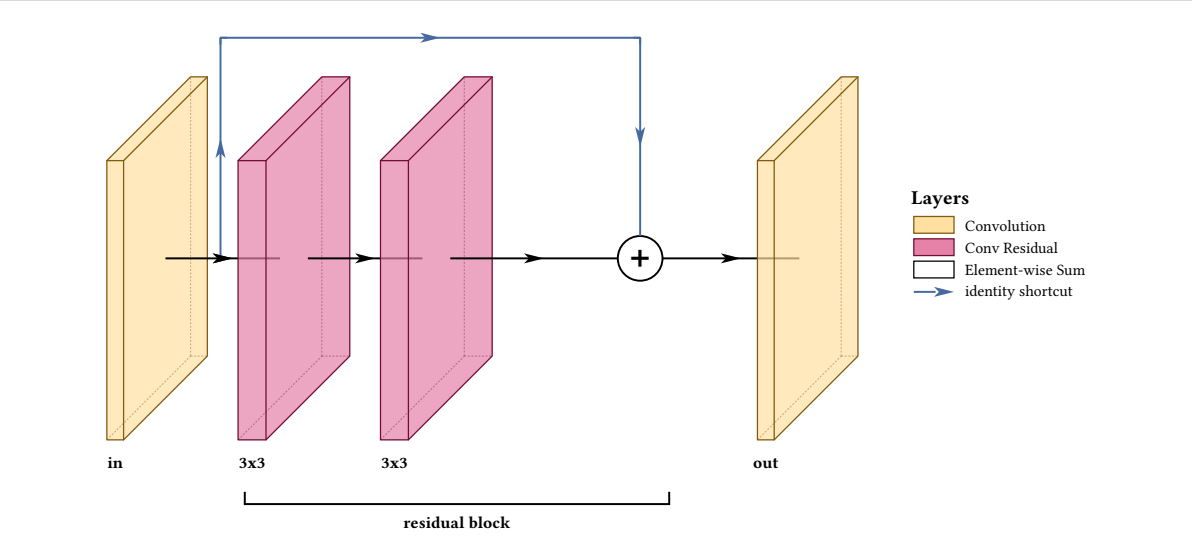
```
#draw-network((
  input(image: "default", show-connection: true),
  conv(label: "conv", offset: 1.5),
))
```



10.9 Two composites

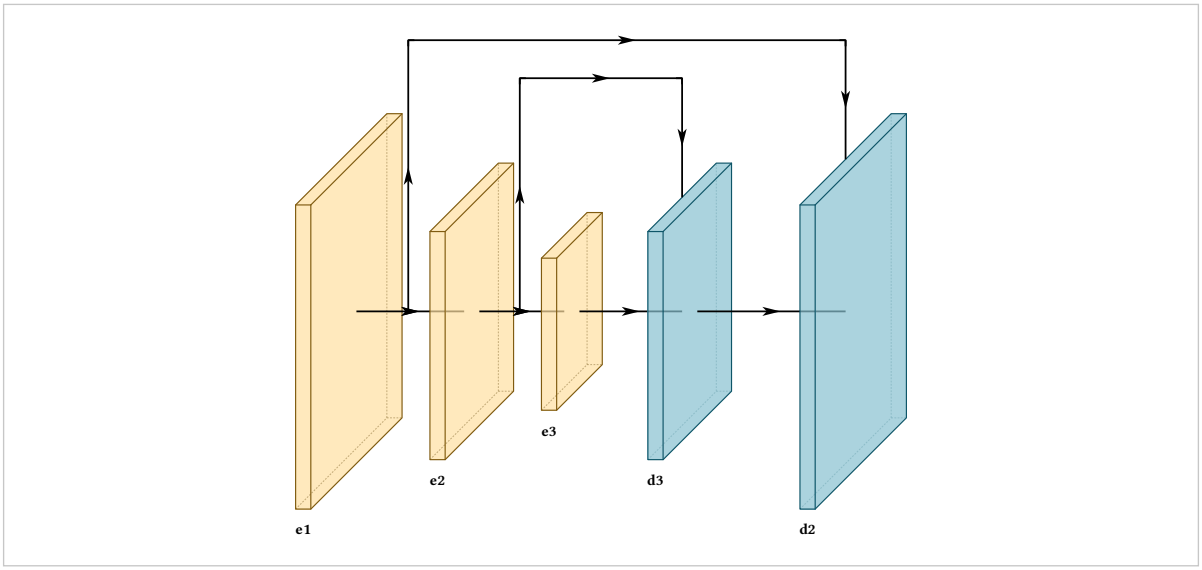
A residual block is a `sum` node plus one route:

```
#draw-network(
  (
    conv(name: "in", label: "in", widths: (0.3,)),
    convres(name: "c1", label: "3x3", widths: (0.5,)),
    convres(name: "c2", label: "3x3", widths: (0.5,)),
    sum(name: "add"),
    conv(name: "out", label: "out", widths: (0.3,)),
  ),
  connections: (
    connection(from: "in", to: "add", color: rgb("#466A9F"),
     legend: "identity shortcut"),
  ),
  groups: (group(from: "c1", to: "add", label: "residual block"),),
  show-legend: true,
)
```



An encoder-decoder is a size pyramid plus two routes arriving on their targets:

```
#let enc = (widths: (0.3,))
#draw-network(
  (
    conv(name: "e1", label: "e1", height: 6, depth: 6, ..enc),
    conv(name: "e2", label: "e2", height: 4.5, depth: 4.5, ..enc),
    conv(name: "e3", label: "e3", height: 3, depth: 3, ..enc),
    deconv(name: "d3", label: "d3", height: 4.5, depth: 4.5),
    deconv(name: "d2", label: "d2", height: 6, depth: 6),
  ),
  connections: (
    connection(from: "e2", to: "d3", touch-layer: true),
    connection(from: "e1", to: "d2", touch-layer: true),
  ),
)
```



CHAPTER 11

Groups and repeats

Backbone, neck and head are the phrases anyone uses out loud to explain one of these figures. `groups` draws them, as labelled brackets under a span of layers.

`from` default `required`

First layer of the span, by name.

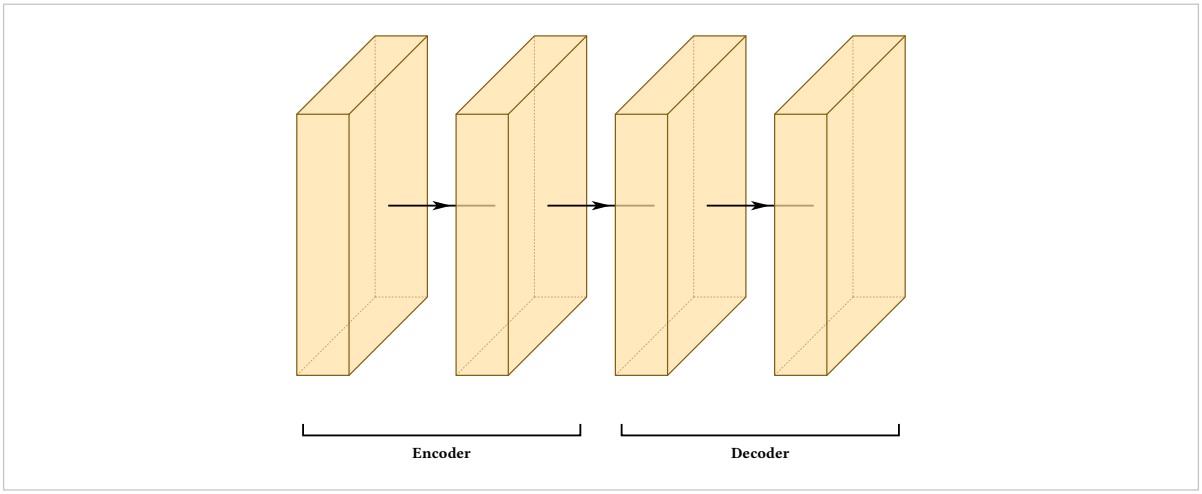
`to` default `required`

Last layer of the span, inclusive. May be the same layer.

`label` default `none`

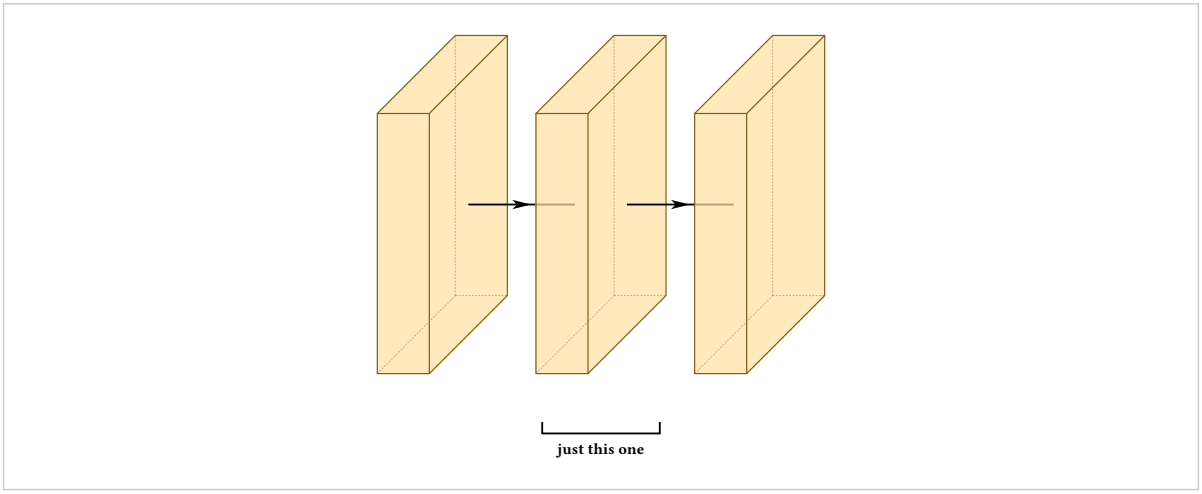
Bracket text.

```
#draw-network(
  (
    conv(name: "e1"), conv(name: "e2"),
    conv(name: "d1"), conv(name: "d2"),
  ),
  groups: (
    group(from: "e1", to: "e2", label: "Encoder"),
    group(from: "d1", to: "d2", label: "Decoder"),
  ),
)
```



A group of one is allowed and common, since a head is often a single block:

```
#draw-network(
  (conv(name: "a"), conv(name: "b"), conv(name: "c")),
  groups: (group(from: "b", to: "b", label: "just this one"),),
)
```



The bracket covers the drawn footprint rather than the front faces, so it sits under the whole block including its lean, and its ends are inset slightly so two adjacent groups read as two rather than as one continuous rule.

11.1 Tinting and stacking

`color`

default `grey`

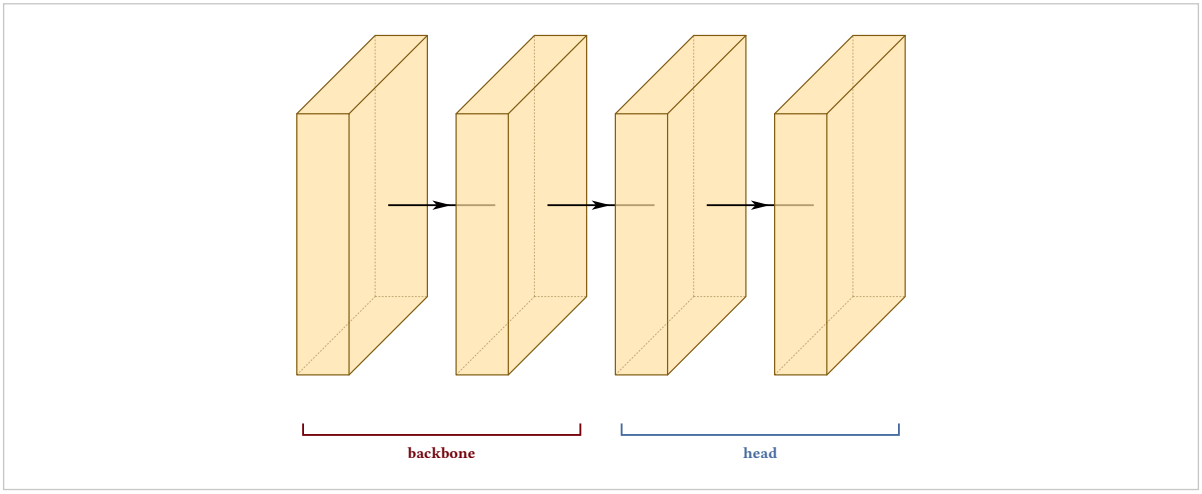
Tints one bracket.

`offset`

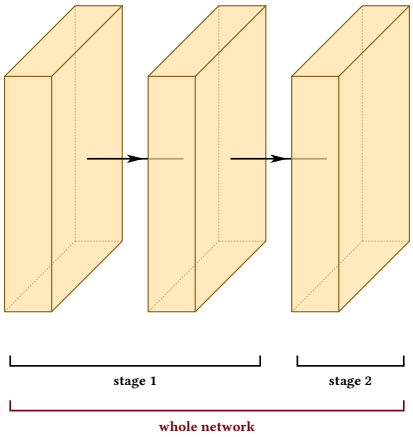
default `1.15`

Pushes the bracket further down, which is how you stack a group that encloses other groups onto its own row.

```
#draw-network(
  (conv(name: "a"), conv(name: "b"), conv(name: "c"), conv(name: "d")),
  groups: (
    group(from: "a", to: "b", label: "backbone", color: rgb("#73000A")),
    group(from: "c", to: "d", label: "head", color: rgb("#466A9F")),
  ),
)
```

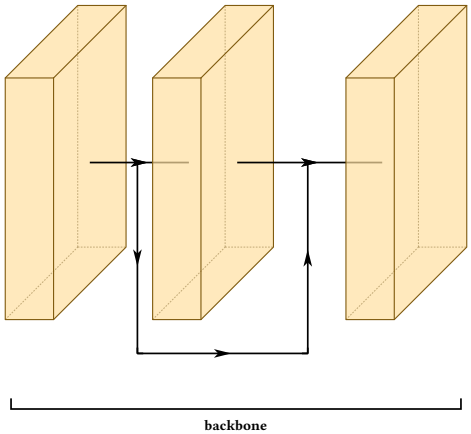


```
#draw-network(
  (conv(name: "a"), conv(name: "b"), conv(name: "c")),
  groups: (
    group(from: "a", to: "b", label: "stage 1"),
    group(from: "c", to: "c", label: "stage 2"),
    group(from: "a", to: "c", label: "whole network",
     offset: 2.1, color: rgb("#73000A")),
  ),
)
```



Brackets are placed below everything else in the figure, including a route running underneath the stack, so the two never collide:

```
#draw-network(
  (conv(name: "a"), conv(name: "b"), conv(name: "c")),
  connections: (connection(from: "a", to: "c", mode: "flat"),),
  groups: (group(from: "a", to: "c", label: "backbone"),),
)
```



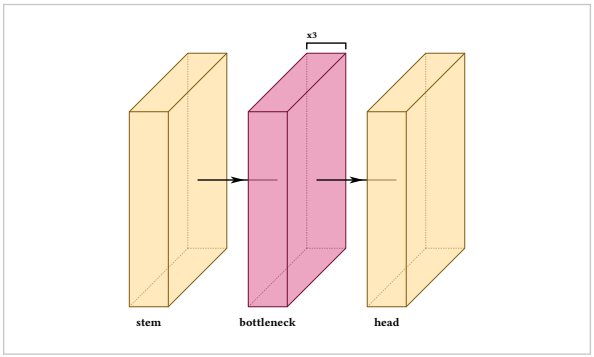
11.2 Repeated blocks

Depth-scaled models stack the same block several times. Writing that as extra entries in `widths` makes it **look** repeated, but the package has no idea it is: it cannot label the repeat or bracket it, and the figure claims one wide block rather than several.

`repeat` default `1`

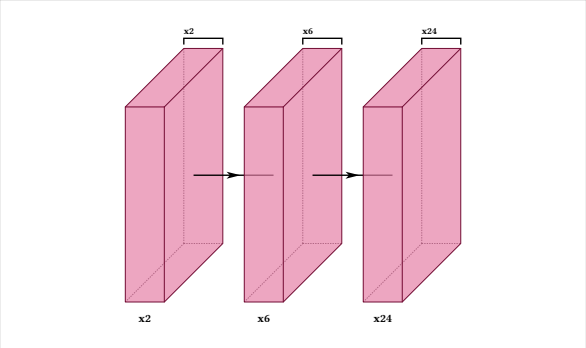
Marks the block as repeated in series. It is drawn once, with a bracket above it carrying the count. Ignored on `pool`, `unpool` and `sum`.

```
#draw-network((
  conv(label: "stem"),
  convres(label: "bottleneck", repeat: 3),
  conv(label: "head"),
))
```



The count costs no horizontal space, so it stays legible at any depth:

```
#draw-network((
  convres(label: "x2", repeat: 2),
  convres(label: "x6", repeat: 6),
  convres(label: "x24", repeat: 24),
))
```

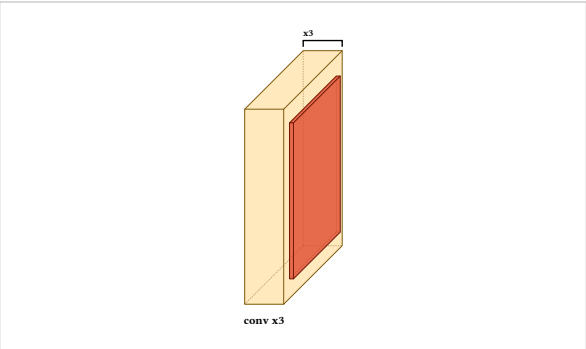


> Ghosted copies were tried first and rejected. An outline behind the block reads as an empty box rather than as another one of the same block, and drawing N of them is either misleading about the count or unreadable once N is large. A bracket states the count instead of depicting it.

The pool pitfall

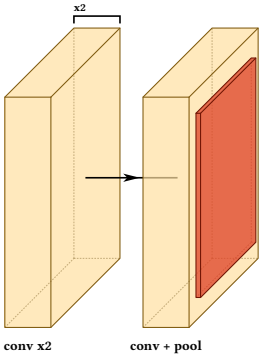
`repeat` describes one layer entry, not a run of them. Putting it on a convolution that has a pool attached reads as though the pool repeats too:

```
#draw-network((
  conv(label: "conv x3", repeat: 3),
  pool(),
))
```



Split it instead. Let the repeat cover the plain convolutions, and draw the last one on its own with the pool attached:

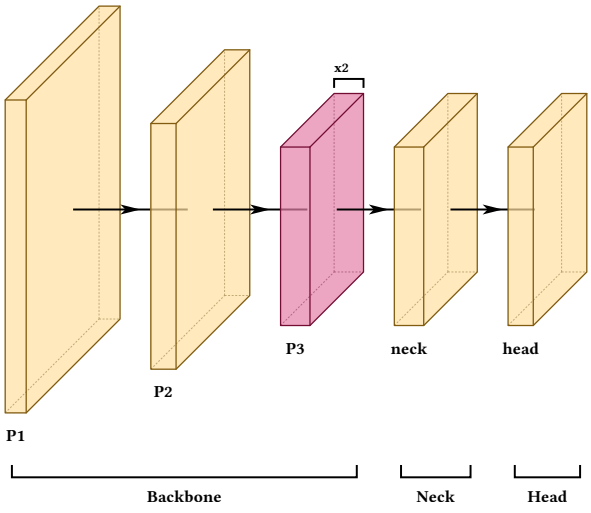
```
#draw-network((
  conv(label: "conv x2", repeat: 2),
  conv(label: "conv + pool", offset: 2),
  pool(),
))
```



For the same reason a repeated **pair**, such as attention-then-MLP, cannot be expressed with `repeat` at all. Section 15.5 shows what to do instead.

11.3 A composite

```
#draw-network(
  (
    conv(name: "s1", label: "P1", shape: (32, 160, 160)),
    conv(name: "s2", label: "P2", shape: (64, 80, 80)),
    convres(name: "s3", label: "P3", shape: (128, 40, 40), repeat: 2),
    conv(name: "n1", label: "neck", shape: (128, 40, 40)),
    conv(name: "h1", label: "head", shape: (64, 40, 40)),
  ),
  groups: (
    group(from: "s1", to: "s3", label: "Backbone"),
    group(from: "n1", to: "n1", label: "Neck"),
    group(from: "h1", to: "h1", label: "Head"),
  ),
)
```



CHAPTER 12

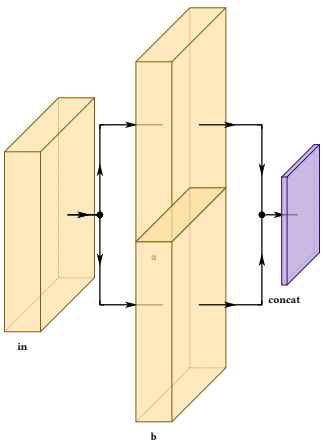
Parallel branches

`draw-network` advances one cursor along one axis, so anything genuinely parallel — two detection heads, the two paths inside a CSP block, a two-stream fusion — would have to be collapsed into a single block with arrows pointed at it. A `branch` entry draws it as it is.

`branches` default `required`

An array of layer arrays, one per parallel path. Each is walked exactly as the trunk is, so anything that works on the trunk works inside a branch.

```
#draw-network((
  conv(label: "in"),
  branch(spread: 5, branches: (
    (conv(label: "a"),),
    (conv(label: "b"),),
  )),
  concat(label: "concat"),
))
```



12.1 Separation and count

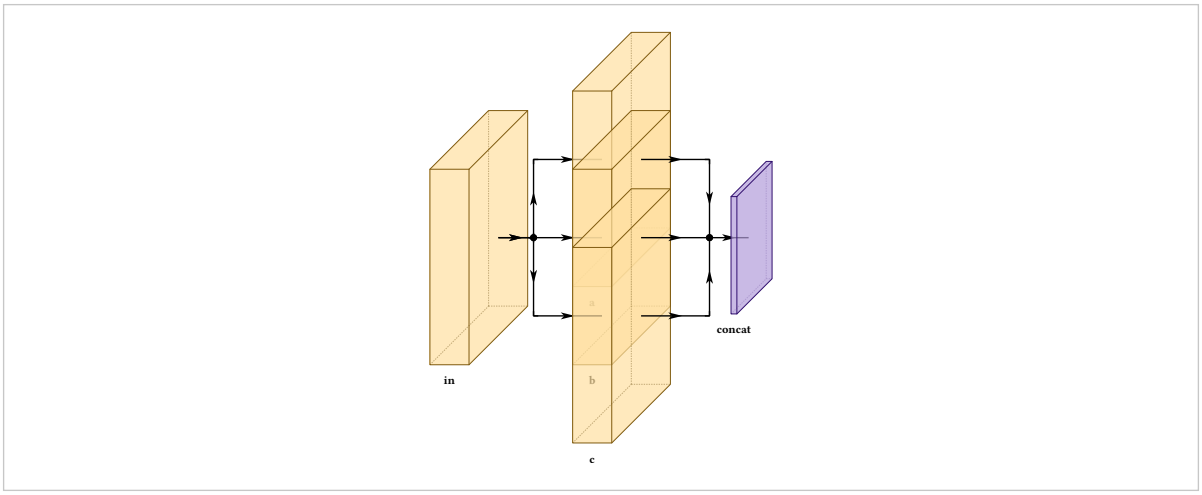
`spread` default `6`

Separation between paths, centred on the trunk. An odd count puts one path on the trunk line; an even count leaves it empty.

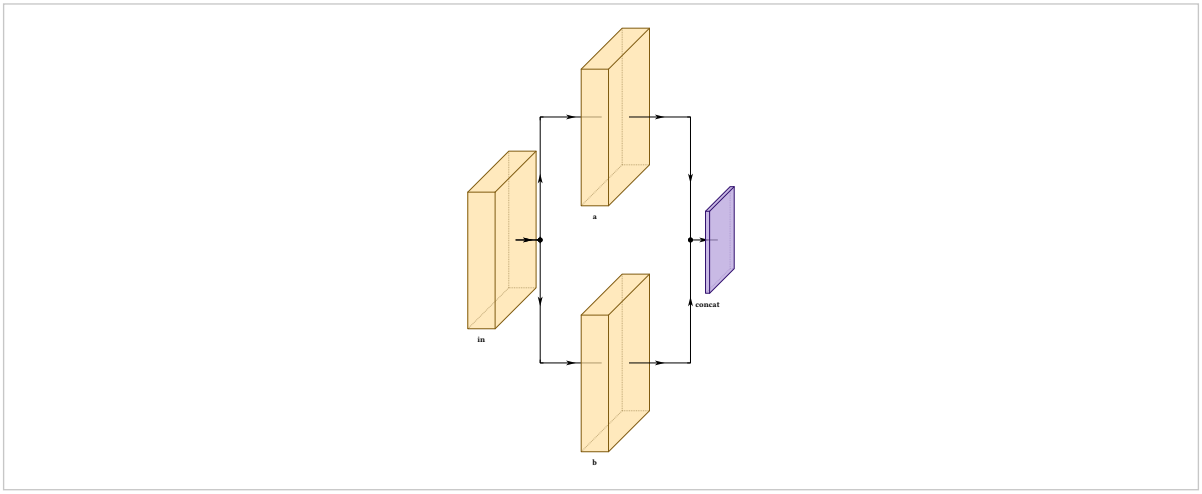
`lead` default `2.0`

How far the fan-out and rejoin arrows run.

```
#draw-network((
  conv(label: "in"),
  branch(spread: 4, branches: (
    (conv(label: "a"),),
    (conv(label: "b"),),
    (conv(label: "c"),),
  )),
  concat(label: "concat"),
))
```

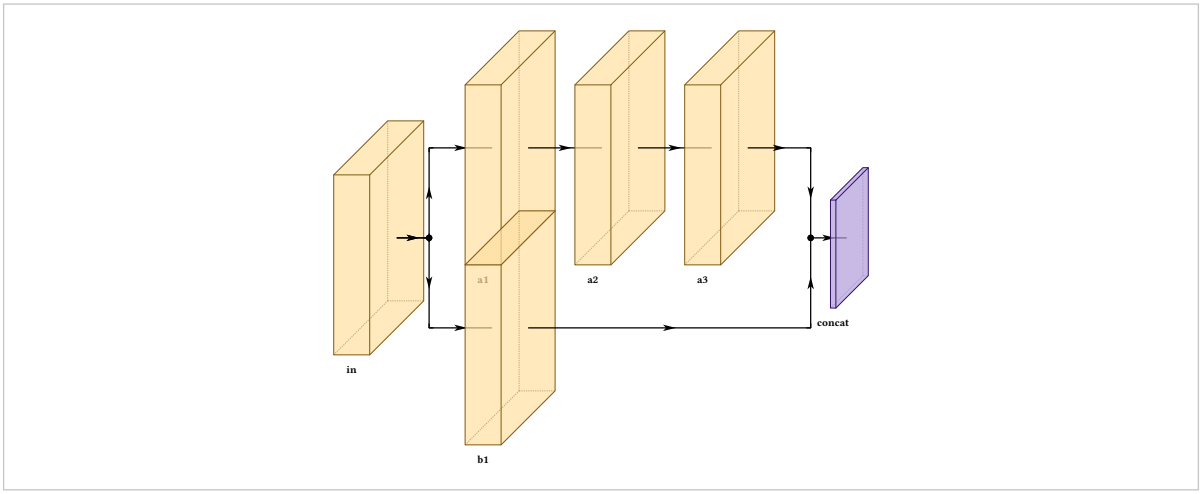



```
#draw-network((
  conv(label: "in"),
  branch(spread: 9, lead: 3, branches: (
    (conv(label: "a"),),
    (conv(label: "b"),),
  )),
  concat(label: "concat"),
))
```



Branches need not be the same length. The rejoin waits for the longest rather than cutting the others short:

```
#draw-network((
  conv(label: "in"),
  branch(spread: 5, branches: (
    (conv(label: "a1"), conv(label: "a2"), conv(label: "a3")),
    (conv(label: "b1"),),
  )),
  concat(label: "concat"),
))
```



A filled dot marks each point where flow divides or meets, and where a tooth leaves a spine that passes through. Merging routes terminate at the dot, and a single arrow leaves it for the next block.

12.2 Stacking along the projection

`spread-mode`

default `"vertical"`

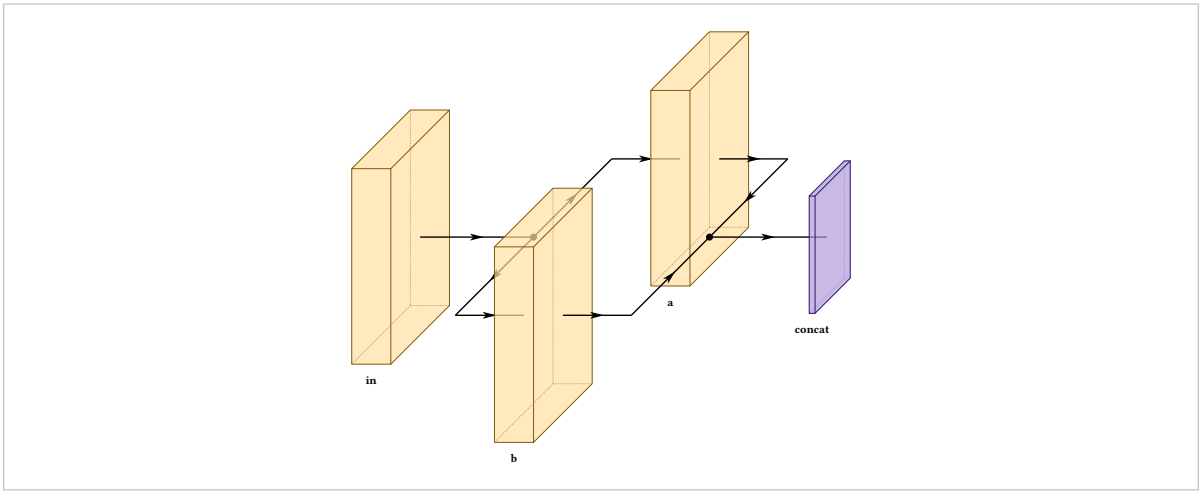
`"depth"` stacks the paths along the projection's 45-degree axis instead of straight up, so they read as parallel copies sitting behind one another.

`rejoin-lead`

default `the value of lead`

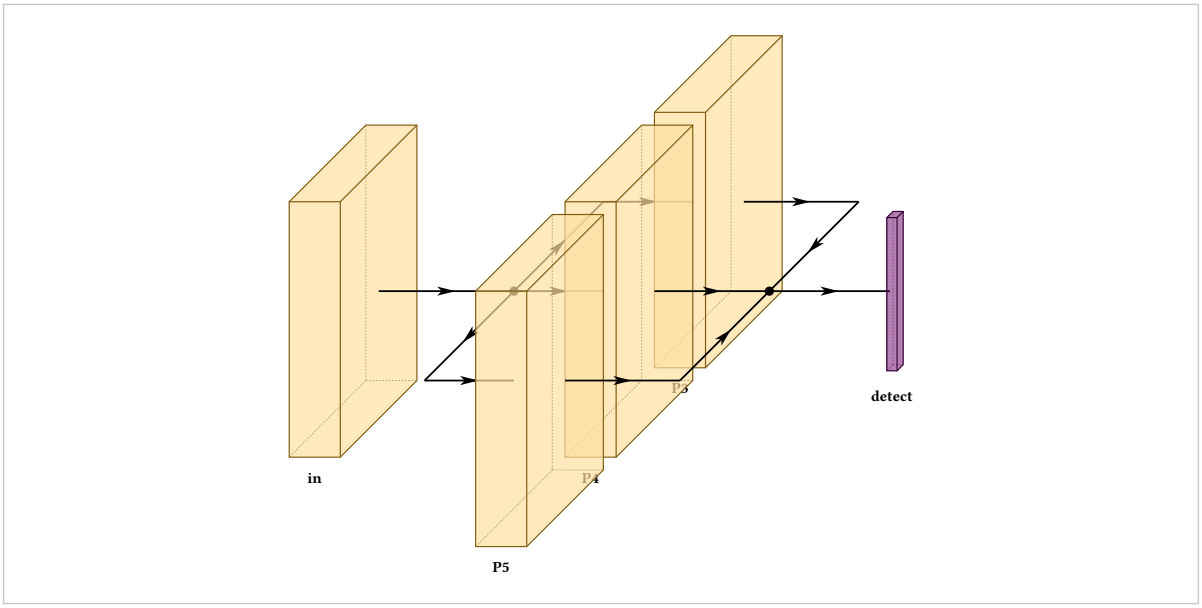
Widens the rejoin side alone. Depth mode often wants this, since its return spine descends past each block's lower-right corner, which is exactly where diagonal dimension labels sit.

```
#draw-network((
  conv(label: "in"),
  branch(spread: 4, spread-mode: "depth", branches: (
    (conv(label: "a"),),
    (conv(label: "b"),),
  )),
  concat(label: "concat"),
))
```



The fan-out and rejoin become two parallel spines with horizontal teeth — a parallelogram, rather than the mirrored hexagon a vertical split produces. Neither 45 is a free angle: it is the one the projection already uses. This is the natural way to draw a set of detection heads:

```
#draw-network((
  conv(label: "in"),
  branch(
    spread: 3.5, spread-mode: "depth", rejoin-lead: 3,
    branches: (
      (conv(label: "P3"),),
      (conv(label: "P4"),),
      (conv(label: "P5"),),
    ),
  ),
  output(label: "detect"),
))
```



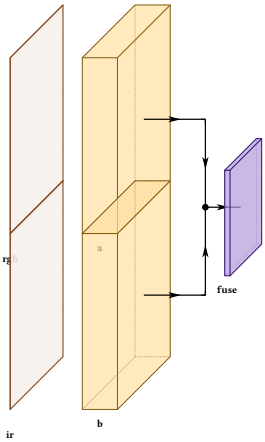
12.3 Open at one end

open

default none

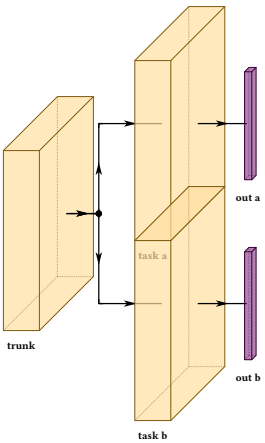
`"start"` draws no fan-out, `"end"` no rejoin. A branch with nothing before it is open at the start by definition. A branch open at the start is how a multi-input network begins:

```
#draw-network((
  branch(spread: 5, branches: (
    (input(label: "rgb"), conv(label: "a")),
    (input(label: "ir"), conv(label: "b")),
  )),
  concat(label: "fuse"),
))
```



Open at the end, one trunk fans out into independent outputs:

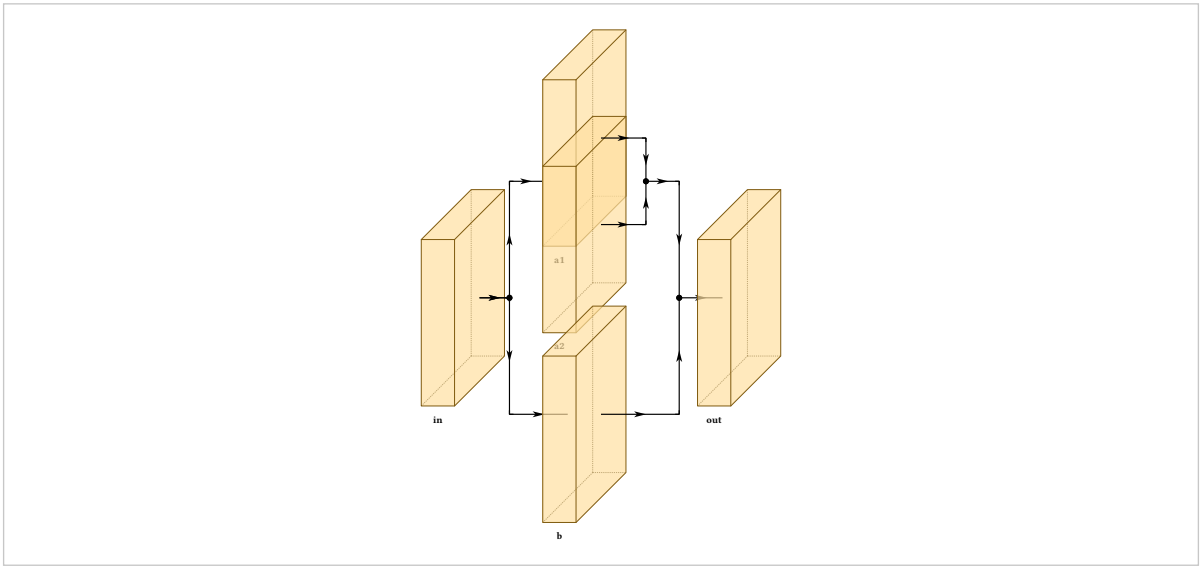
```
#draw-network((
  conv(label: "trunk"),
  branch(spread: 5, open: "end", branches: (
    (conv(label: "task a"), output(label: "out a")),
    (conv(label: "task b"), output(label: "out b")),
  )),
))
```



12.4 Nesting

A branch may contain branches, since walking a branch is the same operation as walking the trunk.

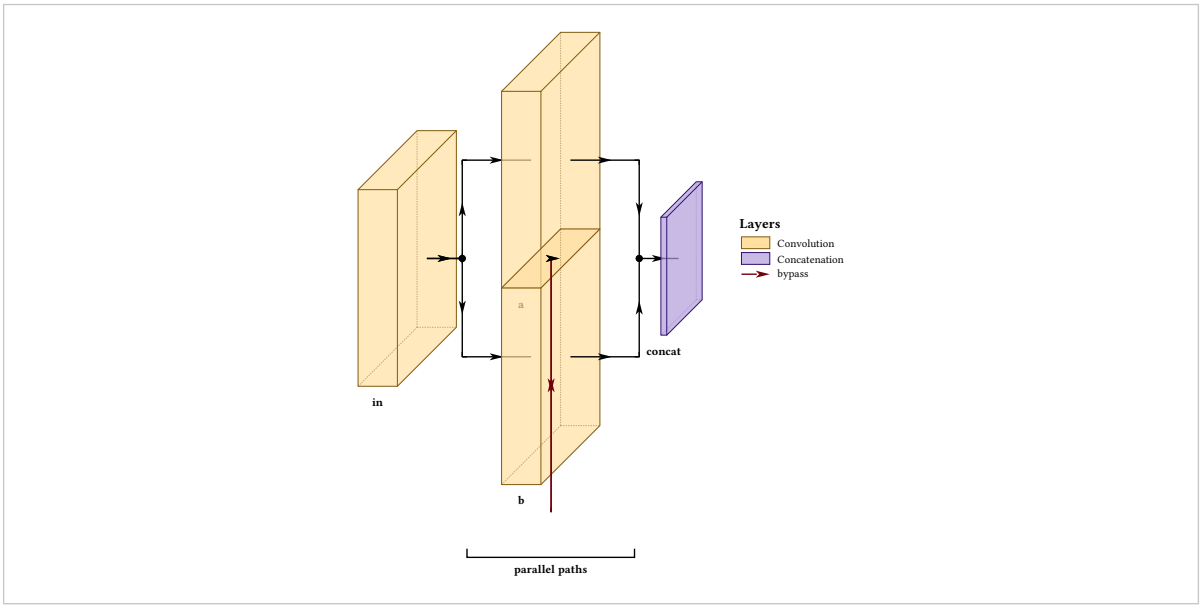
```
#draw-network((
  conv(label: "in"),
  branch(spread: 7, branches: (
    (
      branch(spread: 2.6, lead: 1.2, branches: (
        (conv(label: "a1"),),
        (conv(label: "a2"),),
      )),
    ),
    (conv(label: "b"),),
  )),
  conv(label: "out"),
))
```



12.5 Branches are not a separate world

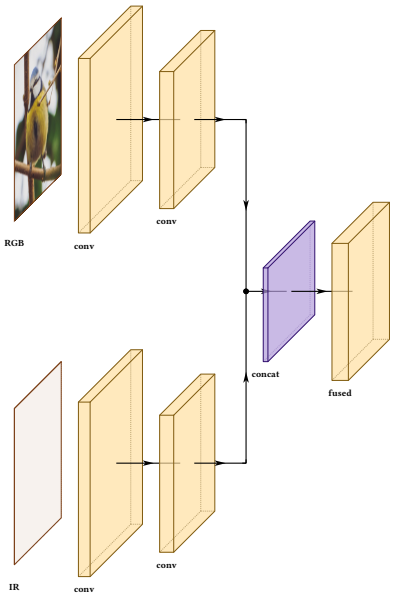
Named layers inside a branch join the same table the trunk uses, so connections, groups and automatic lane ranking all reach into them. A group naming any layer inside a branch widens to the branch's whole drawn extent, plumbing included.

```
#draw-network(
  (
    conv(name: "in", label: "in"),
    branch(spread: 5, branches: (
      (conv(name: "a", label: "a"),),
      (conv(name: "b", label: "b"),),
    )),
    concat(name: "cat", label: "concat"),
  ),
  connections: (
    connection(from: "in", to: "cat", mode: "flat",
     color: rgb("#73000A"), legend: "bypass"),
  ),
  groups: (group(from: "a", to: "b", label: "parallel paths"),),
  show-legend: true,
)
```

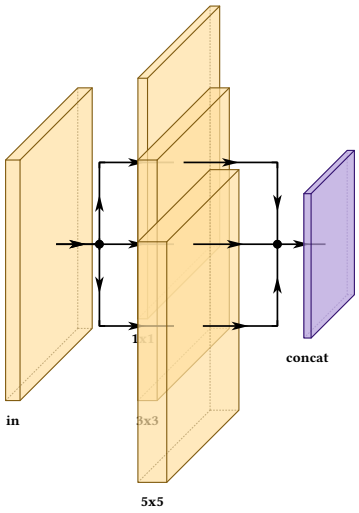


12.6 Two composites

```
#draw-network((
  branch(spread: 11, branches: (
    (
      input(label: "RGB", image: "default"),
      conv(name: "r1", label: "conv", shape: (32, 160, 160)),
      conv(name: "r2", label: "conv", shape: (64, 80, 80)),
    ),
    (
      input(label: "IR"),
      conv(name: "i1", label: "conv", shape: (32, 160, 160)),
      conv(name: "i2", label: "conv", shape: (64, 80, 80)),
    ),
  )),
  concat(label: "concat", depth: 5),
  conv(label: "fused", shape: (128, 80, 80)),
))
```



```
#draw-network((
  conv(label: "in", widths: (0.3,)),
  branch(spread: 3.4, lead: 1.6, branches: (
    (conv(label: "1x1", widths: (0.2,)),),
    (conv(label: "3x3", widths: (0.4,)),),
    (conv(label: "5x5", widths: (0.6,)),),
  )),
  concat(label: "concat"),
))
```



CHAPTER 13

The figure as a whole

13.1 Legends

`show-legend`

default `false`

`draw-network`

Collects every layer type used into a legend, in order of first appearance.

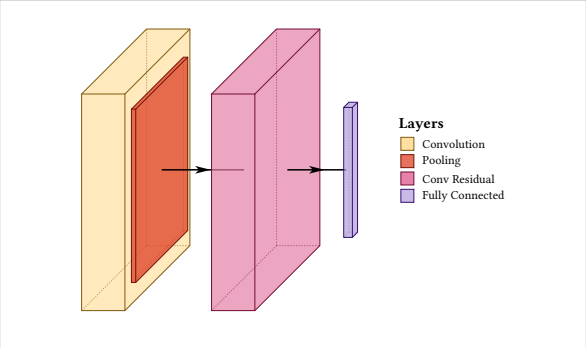
`legend-title`

default `"Layers"`

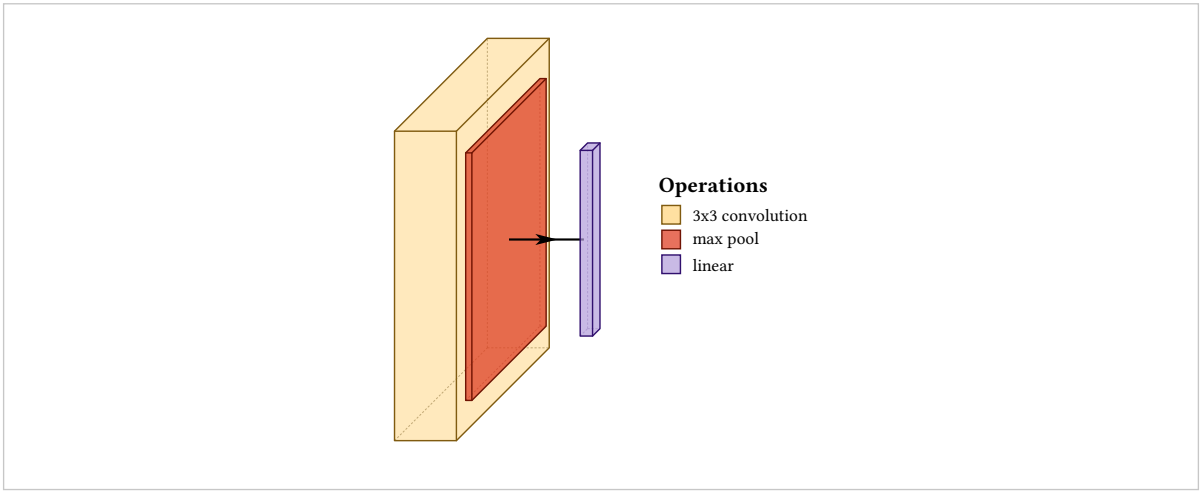
`draw-network`

Heading of the box.

```
#draw-network(
  (conv(), pool(), convres(), fc()),
  show-legend: true,
)
```

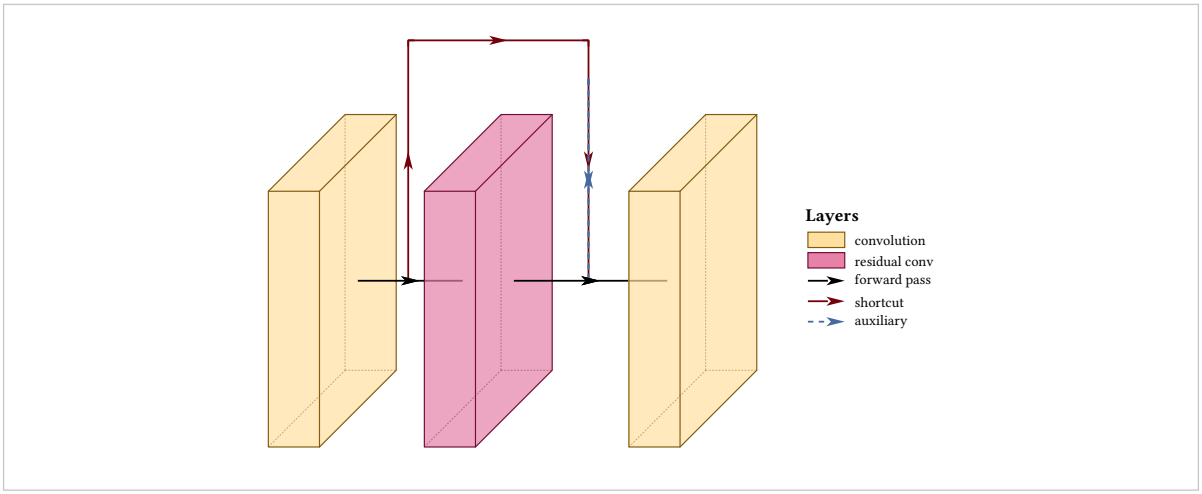



```
#draw-network(
  (
    conv(legend: "3x3 convolution"),
    pool(legend: "max pool"),
    fc(legend: "linear"),
  ),
  show-legend: true,
  legend-title: "Operations",
)
```



Layer swatches and route samples share one box. The sample column widens when any route entry is present, so a dash pattern has room to read, and the layer swatches widen with it so nothing floats away from its label:

```
#draw-network(
  (
    conv(name: "a", legend: "convolution"),
    convres(name: "b", legend: "residual conv"),
    conv(name: "c"),
  ),
  connections: (
    connection(from: "a", to: "c", color: rgb("#73000A"), legend: "shortcut"),
    connection(from: "b", to: "c", color: rgb("#466A9F"), dash: "dashed",
     legend: "auxiliary"),
  ),
  show-legend: true,
  main-legend: "forward pass",
)
```



13.2 Fitting the page

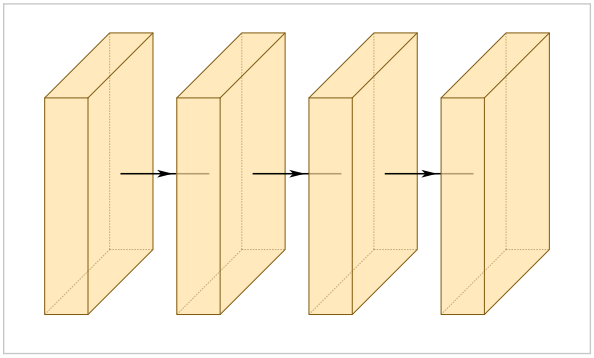
`scale`

default `100%`

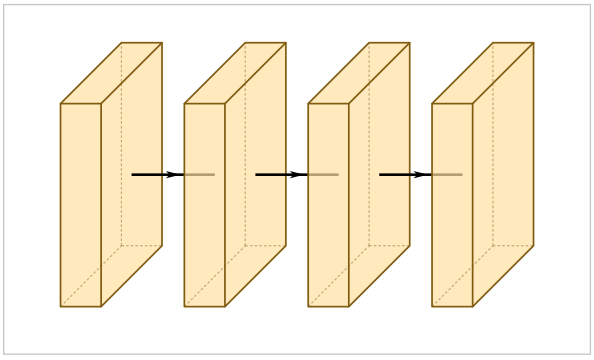
draw-network

Shrinks or grows the whole figure, text included.

```
#draw-network(
  (conv(), conv(), conv(), conv()),
  scale: 100%,
)
```



```
#draw-network(
  (conv(), conv(), conv(), conv()),
  scale: 55%,
)
```



A long network is wider than a page at its natural size, and `scale` is the first thing to reach for. If it makes the text too small to read, the next thing to try is `label-orient: "diagonal"` (section 6.5), which usually buys back enough horizontal space to raise the scale again.

13.3 Line weight

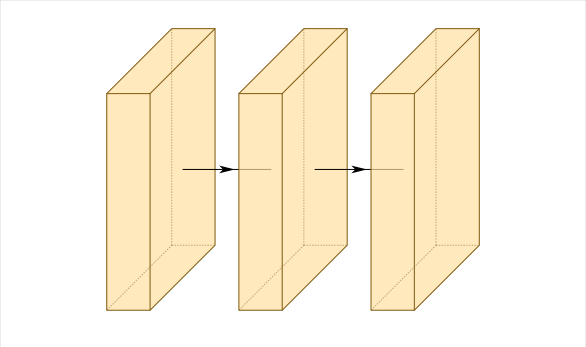
`stroke-thickness`

default `1`

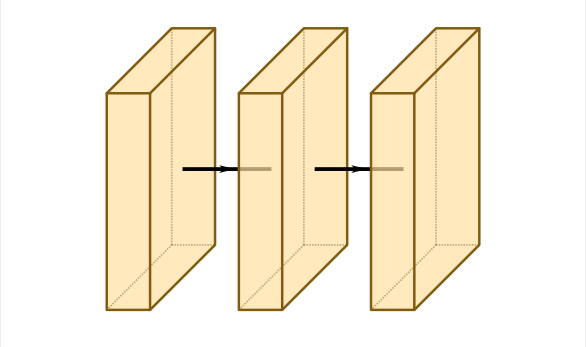
draw-network

Multiplies every stroke in the figure. Connection `thickness` multiplies on top of this.
Worth raising when a figure is reduced for a two-column layout, where hairlines start dropping out at print resolution:

```
#draw-network(
  (conv(), conv(), conv()),
  stroke-thickness: 0.5,
)
```



```
#draw-network(
  (conv(), conv(), conv()),
  stroke-thickness: 2.5,
)
```



13.4 The projection

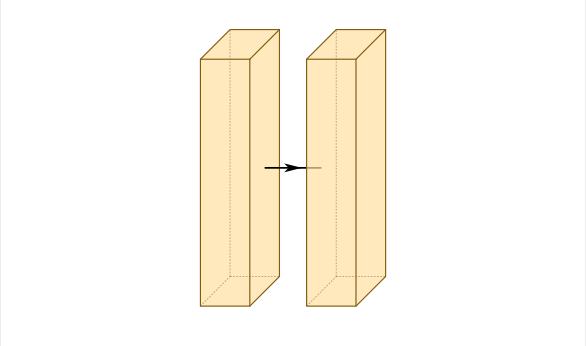
`depth-multiplier`

default `0.3`

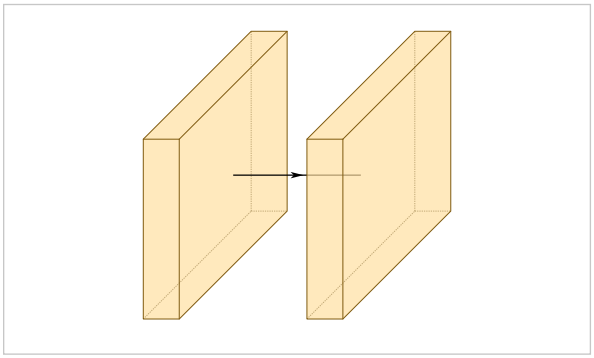
draw-network

How far blocks lean, which is the strength of the isometric projection. It also changes what `depth-shear` and `min-clear-offset` return, so pass the same value to those.

```
#draw-network(
  (conv(depth: 5), conv(depth: 5)),
  depth-multiplier: 0.12,
)
```



```
#draw-network(
  (conv(depth: 5), conv(depth: 5)),
  depth-multiplier: 0.6,
)
```



CHAPTER 14

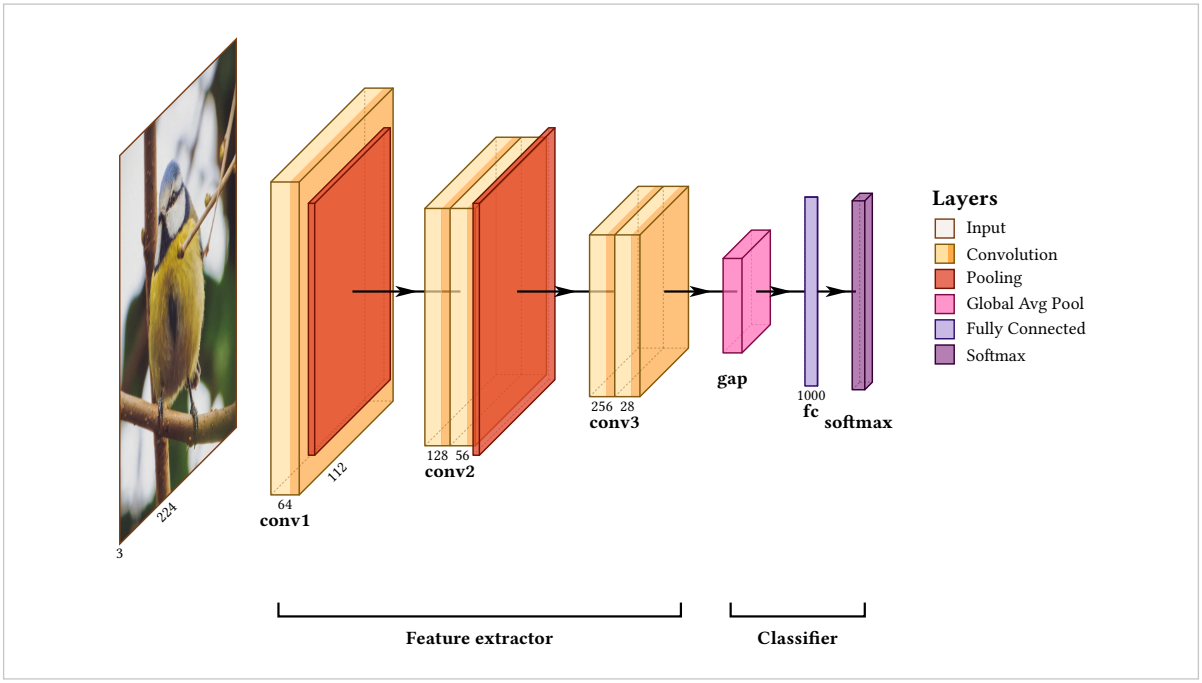
Patterns

Everything here is built only from what the preceding chapters covered. These are not architectures to copy so much as shapes to recognise: most figures are one of them with different labels.

14.1 A classifier

Types, shapes and two groups.

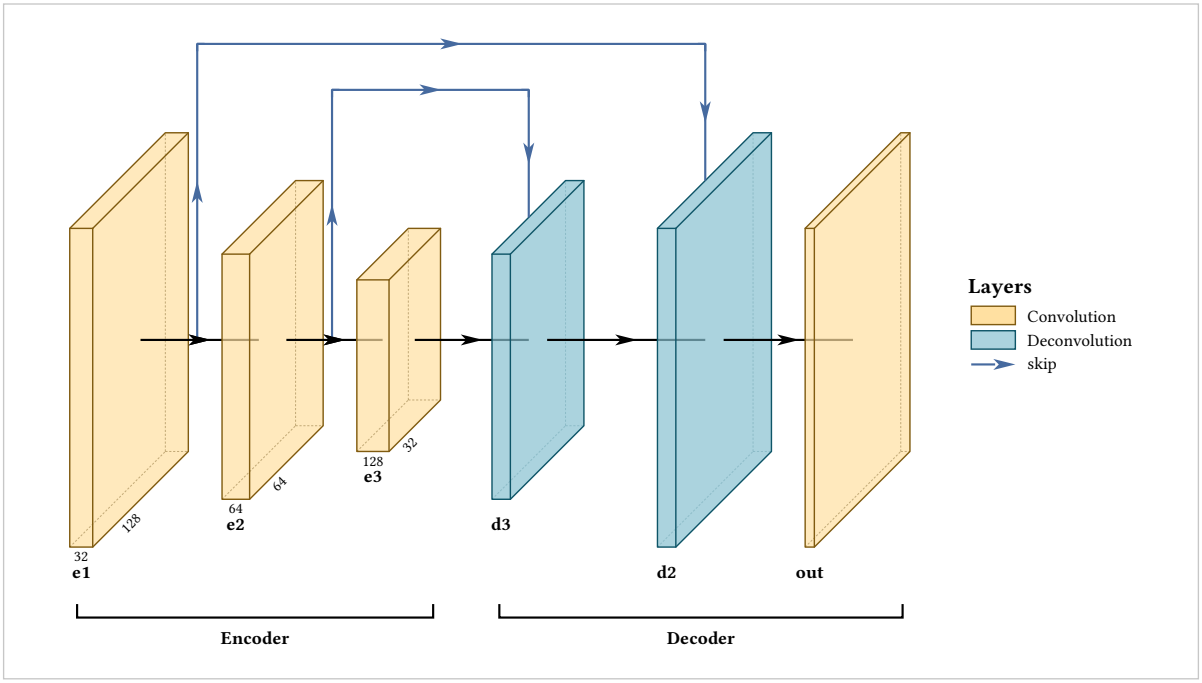
```
#draw-network(
  (
    input(name: "in", image: "default", shape: (3, 224, 224),
          channels: (3, 224)),
    conv(name: "c1", label: "conv1", shape: (64, 112, 112),
         channels: (64, 112)),
    pool(),
    conv(name: "c2", label: "conv2", shape: (128, 56, 56),
         channels: (128, 56), widths: (0.4, 0.4)),
    pool(),
    conv(name: "c3", label: "conv3", shape: (256, 28, 28),
         channels: (256, 28), widths: (0.4, 0.4)),
    gap(name: "g", label: "gap"),
    fc(name: "f", label: "fc", channels: (1000,), depth: 0),
    softmax(name: "s", label: "softmax"),
  ),
  groups: (
    group(from: "c1", to: "c3", label: "Feature extractor"),
    group(from: "g", to: "s", label: "Classifier"),
  ),
  show-legend: true,
  show-relu: true,
)
```



14.2 An encoder-decoder

A size pyramid down and back up, with skips arriving on their targets. The two `deconv` blocks take their size from `shape` exactly as the encoder does, so the figure is symmetric because the model is, not because it was measured.

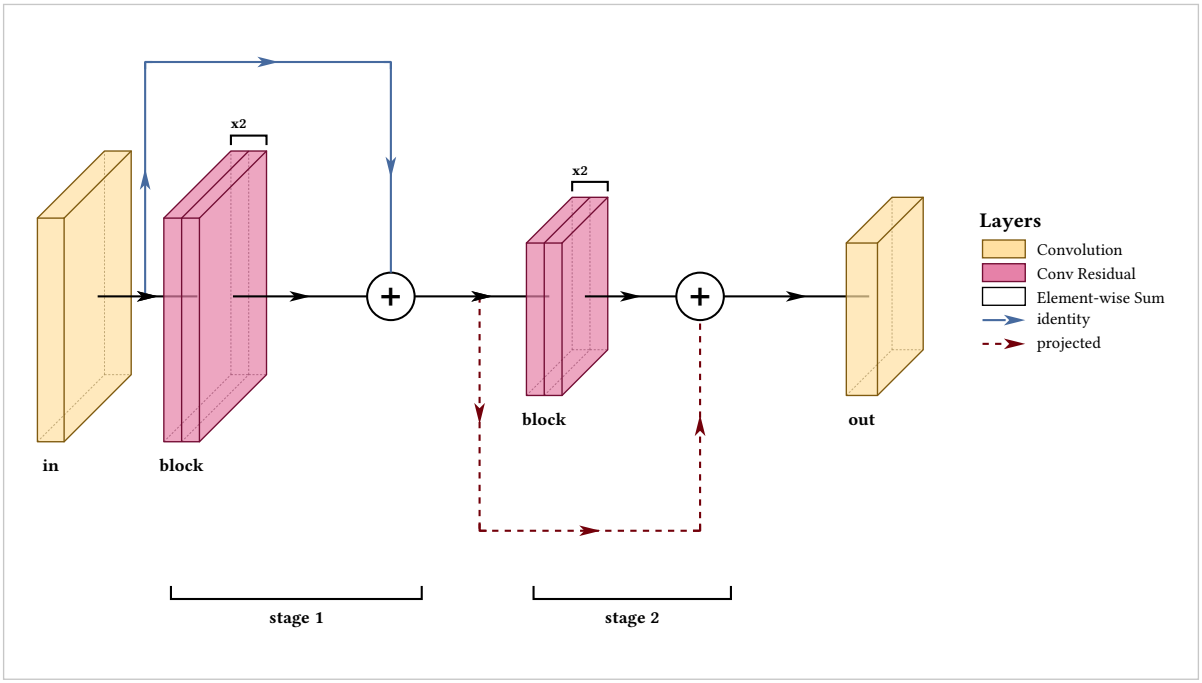
```
#let e(n, s) = conv(name: n, label: n, shape: s, channels: (s.at(0), s.at(1)))
#let d(n, s) = deconv(name: n, label: n, shape: s)

#draw-network(
  (
    e("e1", (32, 128, 128)),
    e("e2", (64, 64, 64)),
    e("e3", (128, 32, 32)),
    d("d3", (64, 64, 64)),
    d("d2", (32, 128, 128)),
    conv(name: "out", label: "out", shape: (1, 128, 128)),
  ),
  connections: (
    connection(from: "e2", to: "d3", touch-layer: true,
     color: rgb("#466A9F"), legend: "skip"),
    connection(from: "e1", to: "d2", touch-layer: true,
     color: rgb("#466A9F")),
  ),
  groups: (
    group(from: "e1", to: "e3", label: "Encoder"),
    group(from: "d3", to: "out", label: "Decoder"),
  ),
  show-legend: true,
)
```



14.3 Residual stages

`repeat` for the blocks inside a stage, a `sum` node for each add, and two route colours to separate an identity shortcut from a projected one.

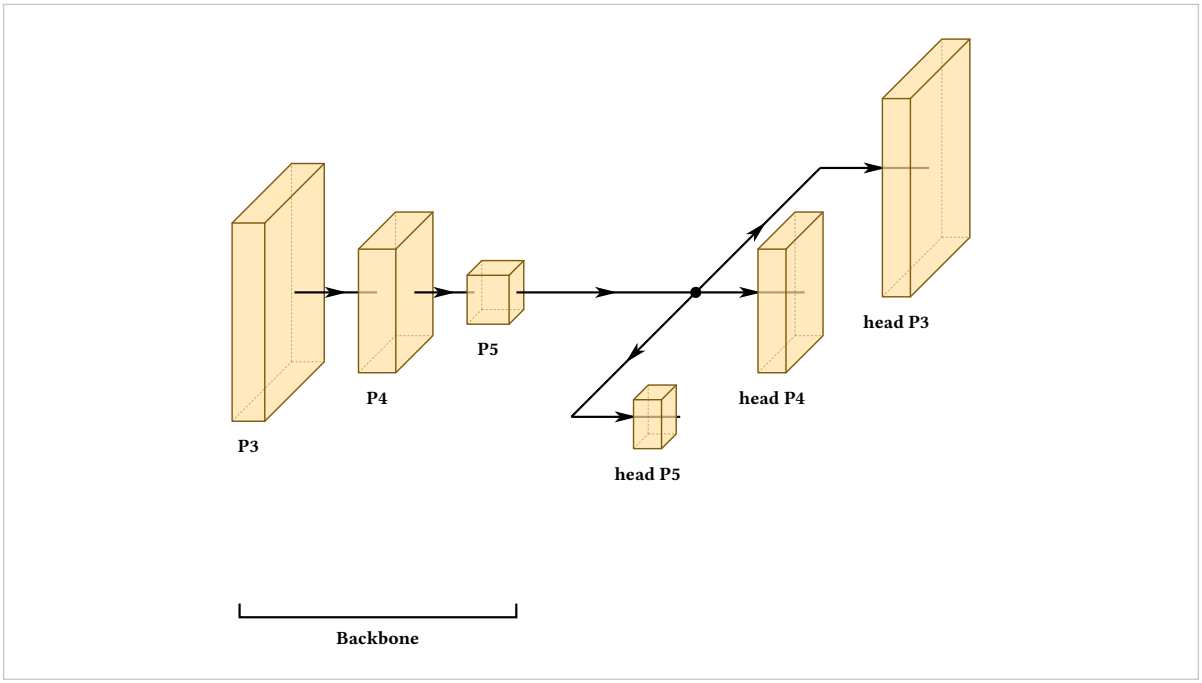
```
#draw-network(
  (
    conv(name: "in", label: "in", shape: (64, 56, 56)),
    convres(name: "b1", label: "block", shape: (64, 56, 56),
            widths: (0.3, 0.3), repeat: 2),
    sum(name: "a1"),
    convres(name: "b2", label: "block", shape: (128, 28, 28),
            widths: (0.3, 0.3), repeat: 2),
    sum(name: "a2"),
    conv(name: "out", label: "out", shape: (128, 28, 28)),
  ),
  connections: (
    connection(from: "in", to: "a1", color: rgb("#466A9F"),
     legend: "identity"),
    connection(from: "a1", to: "a2", color: rgb("#73000A"), dash: "dashed",
     legend: "projected"),
  ),
  groups: (
    group(from: "b1", to: "a1", label: "stage 1"),
    group(from: "b2", to: "a2", label: "stage 2"),
  ),
  show-legend: true,
)
```



14.4 A multi-scale head

A depth-mode branch, open at the end, is what a detector's heads look like.

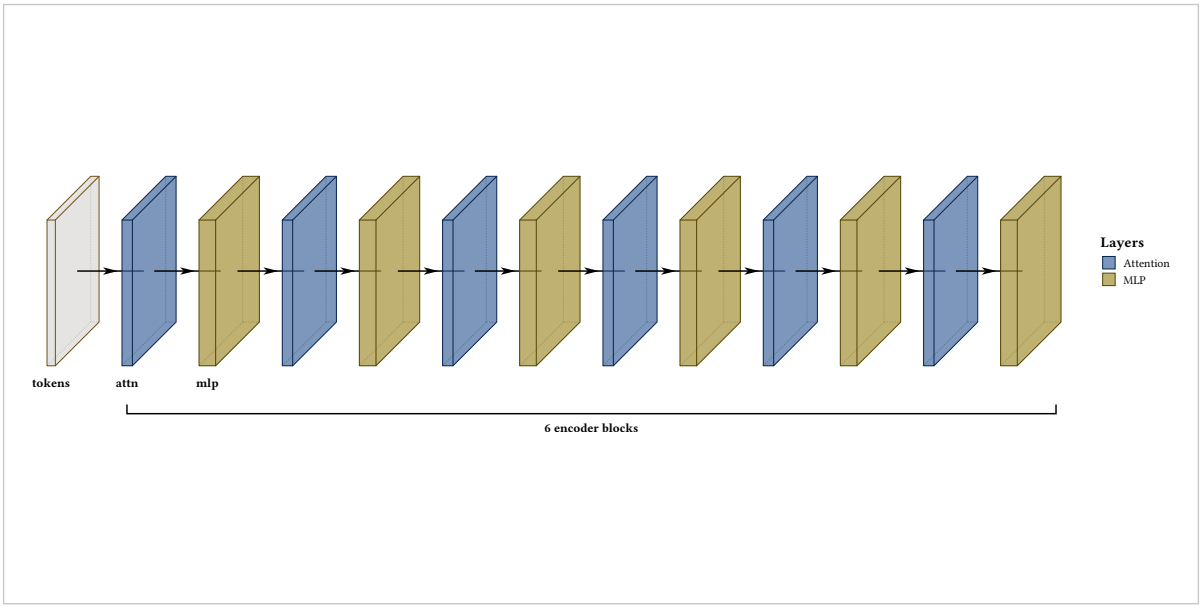
```
#draw-network(
  (
    conv(name: "p3", label: "P3", shape: (128, 40, 40)),
    conv(name: "p4", label: "P4", shape: (256, 20, 20)),
    conv(name: "p5", label: "P5", shape: (512, 10, 10)),
    branch(spread: 4, spread-mode: "depth", rejoin-lead: 3.2,
      open: "end", branches: (
        (conv(name: "h3", label: "head P3", shape: (64, 40, 40)),),
        (conv(name: "h4", label: "head P4", shape: (64, 20, 20)),),
        (conv(name: "h5", label: "head P5", shape: (64, 10, 10)),),
      )),
  ),
  groups: (group(from: "p3", to: "p5", label: "Backbone"),),
)
```



14.5 A repeated pair

`repeat` cannot express a repeated **pair** (section 11.3), so build the list instead. Twelve blocks are written once, and the count lives in the `range` rather than in the figure:

```
#let block(i) = (
  custom(name: "a" + str(i), label: if i == 1 { "attn" },
         fill: rgb("#466A9F"), width: 0.25, height: 3.5, depth: 3.5,
         legend: "Attention"),
  custom(name: "m" + str(i), label: if i == 1 { "mlp" },
         fill: rgb("#A49137"), width: 0.4, height: 3.5, depth: 3.5,
         legend: "MLP"),
)

#draw-network(
  (custom(name: "in", label: "tokens", width: 0.2,
          height: 3.5, depth: 3.5),)
    + range(1, 7).map(block).flatten(),
  groups: (group(from: "a1", to: "m6", label: "6 encoder blocks"),),
  show-legend: true,
)
```



This is the general answer to any figure that is getting long. The layer list is an array and a constructor is a function; anything Typst can do to those, you can do to a figure.

CHAPTER 15

Every option

The tables below are generated from the package itself rather than written out here, so an option cannot exist without appearing in this chapter.

15.1 Options by layer type

Type	Options
branch	branches, lead, name, open, rejoin-lead, spread, spread-mode
concat	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
conv	bandfill, channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, show-relu, widths, xlabel, ylabel, zlabel
convres	bandfill, channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, show-relu, widths, xlabel, ylabel, zlabel
convsoftmax	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
custom	bandfill, channels, connection-label, depth, fill, height, image, input-style, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, show-relu, width, widths, xlabel, ylabel, zlabel
deconv	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
fc	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection
gap	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection
input	channels, connection-label, depth, fill, height, image, input-style, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
output	channels, classes, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
pool	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection
softmax	channels, classes, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
sum	channels, connection-label, depth, fill, height, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, radius, show-connection, stroke, symbol
unpool	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection

15.2 Connection options

arrive-offset · clearance · color · dash · from · label · layers · legend · mode · opacity · pos · thickness · to · touch-layer · type

15.3 Group options

`color · from · label · offset · to`

15.4 Arguments to draw-network

```
#draw-network(
  layers,
  connections: (),
  groups: (),
  palette: "warm",
  show-legend: false,
  legend-title: "Layers",
  main-legend: none,
  scale: 100%,
  stroke-thickness: 1,
  depth-multiplier: 0.3,
  lane-unit: 0.75,
  shape-scale: (spatial: (1.2, -3.2), channels: (0.075, 0.0)),
  auto-gap: 0.55,
  show-relu: false,
)
```

15.5 Also exported

```
depth-shear(depth, depth-multiplier: 0.3)
min-clear-offset(depth, depth-multiplier: 0.3)
```

Plus a constructor per layer type, and `connection(..)` and `group(..)`.

15.6 Coverage

> Every option exported by the package is described somewhere above.

CHAPTER 16

Errors

Every message the package raises, and what causes it.

`unknown layer option "X" on layer N (type "T")`

`X` is not an option of that type. The message lists what is, and suggests the nearest match. A correctly spelled option belonging to a **different** type gives this too — `widths` on a `pool`, `width` on a `conv`.

`unknown layer type "X" on layer N`

Not one of the fourteen. See section 16.1.

`layer N is a plain dictionary`

Every layer comes from a constructor. Write `(type: "conv", ..)` as `conv(..)`; a dictionary of options spreads into it as `conv(..opts)`.

`connection N refers to "X", which is not the `name` of any layer`

Either the target has no `name`, or the reference is misspelled. Groups give the same message.

`connection N has no `from``

A route names both layers it runs between.

``branches` on layer N must be an array of layer arrays`

A common slip is passing one layer list rather than a list of them. Two branches of one layer each is `((conv(),), (conv(),))`.

`branch open must be "start" or "end"`

`label-orient must be "horizontal", "diagonal" or "vertical"`

`Use either label-orient or label-angle, not both`

The right anchor for an arbitrary rotation depends on where you want the text, so the package will not guess.

`shape must be (channels, height, width)`

Three numbers, in that order.

`array index out of bounds (index: 1, len: 1)`

Raised from inside `src/lib.typ` rather than as a message of the package's own. Almost always `channels` with more entries than the layer has bands, plus one. Either add matching `widths` entries or remove channel entries.

`unexpected argument: X`

Typst's own message, from a constructor given an option that does not exist. It is raised before the package sees the call, which is one reason to prefer constructors.

CHAPTER 17

Where to go next

Complete architectures are in `examples/` in the repository, and their rendered output in `gallery/`. They are ordinary Typst files, meant to be read and copied:

`examples/networks/` AlexNet, LeNet-5, VGG16, VGG19, ResNet18, U-Net, FCN-8 and YOLO26-n, plus five RGB-IR fusion variants of the last one.

`examples/features/` every predefined layer type side by side in both palettes, and a figure exercising every feature at once.

`examples/imported/` figures generated from a traced PyTorch model rather than written by hand, using `tools/import_model.py`.

The YOLO26-n example is the one to read if you want to see the automatic machinery carrying a whole figure. Every block states its tensor shape, every gap is automatic, every lane height is automatic, and the three detection heads are a depth-mode branch. Nothing in it is sized or positioned by hand.

This manual is itself plain text. `doc/manual.typ` and the 125 files in `doc/examples/` are the whole of it, each example a standalone figure that compiles on its own. That source, rather than this PDF, is what to point a language model at.